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ABSTRACT

In this deliverable we provide an update of the application requirements. The initial requirement specification has been described as part of deliverable D2.1. Two prototypes of the main applications that drive the design of the CPaaS.io platform have been described in deliverables D2.2 and D2.4 respectively. As expected, experiences have been gained while developing the application and receiving feedback from stakeholders both from an application as well as from a platform perspective. In addition, we made progress towards the federation of the European and Japanese platform and the transferability of applications as described in deliverable D2.7. We have also collected data at the Sapporo Snow Festival that we consider for integration in the Enhanced User Experience application. These developments are reflected in this document by providing an update on the scope of the applications and their requirements. We cover all main driving applications plus an additional Smart Parking application that emerged as a particularly suitable application for transferring knowledge to other cities. Other main driving applications include the Sapporo Visitor Experience, Tokyo Public Transportation, Tokyo Management of Service Vehicles, Yokosuka Emergency Medical Care and Waterproof Amsterdam.

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Table of Contents

1	Introduction	7
2	Event Management - Enhanced User Experience	9
2.1	Application Description	9
2.2	Updated Scope	9
2.3	Core Functionalities and Components	10
2.4	Data Sets	11
2.5	Deployment Plan	11
2.6	Requirements	12
3	Event Management – Sapporo Visitor Experience	17
3.1	Application Description	17
3.2	Architecture	17
3.3	User Interface	18
3.4	Data Sets	18
3.5	Deployment Plan	20
3.6	Requirements	20
4	Event Management – Tokyo Public Transportation	25
4.1	Application Description	25
4.2	Architecture	25
4.3	User Interface	25
4.4	Data Sets	25
4.5	Deployment Plan	26
4.6	Requirements	27
5	Event Management – Tokyo Management of Service Vehicle	35
5.1	Application Description	35
5.2	Architecture	36
5.3	User Interface	36
5.4	Data Sets	36
5.5	Deployment Plan	37
5.6	Requirements	38
6	Waterproof Amsterdam	44
6.1	Application description	44
6.2	Updated Scope	45
6.3	Core functionality and component	46
6.4	Data set	47
7.1	Deployment plan	48

7.2	Requirements.....	49
8	Yokosuka Emergency Medical Care.....	53
8.1	Application Description.....	53
8.2	Architecture.....	53
8.3	User Interface	53
8.4	Data Sets.....	53
8.5	Deployment Plan.....	54
8.6	Requirements.....	54
9	Smart Parking.....	60
9.1	Application Description.....	61
9.2	Core Functionalities and Components.....	61
9.3	Data Sets.....	62
9.4	Deployment Plan.....	65
9.5	Requirements.....	65
10	Summary.....	68
	Appendix A – Description of the fields used in the data set description	69
	Appendix B –Description of the fields in the requirements tables.....	70

Table of Acronyms

Acronym	Definition
API	Application Programming Interface
TTN	The Things Network
LoRa	Short for LoRaWAN: Long-Range Wide Area Network
ODPT	Association of Open Data for Public Transportation
OGD	Open Governmental Data
POC	Proof-of-Concept
RPZ	Regulated Parking Zone
UI	User Interface

1 Introduction

This deliverable is the second version of the requirements specification of the use cases developed on top of the CPaaS.io platform as initially described in D2.1. In this deliverable, we provide an updated requirement specification of the use cases required for the second implementation and validation phase. The second phase particularly focuses on the federation capability of the platform. For each application we provide an update of the

- Use case description
- Scope of the application
- Description of core functionalities, components and architecture
- Definition of data sources and data sets
- Description the requirements

For the actual description of the requirements we highlight whether the requirement is old, new or updated. The updated list of requirements is the result of experiences gained while developing the applications (see deliverables D2.2 and D2.4 for a description of the first and second application prototypes) and receiving feedback from stakeholders, both regarding the respective applications as well as from a platform perspective. In particular, we have taken care to reflect scalability and privacy aspects and to link the requirements to the project objectives as described on our web page². These overall project objectives are:

- **Objective 1: Develop an Open Social City Platform.** Open in the sense of Open Data, the platform is accessible to any interested party who then can build value-added applications and services on top of it. Open also in the sense of Open Innovation, additional data sources like sensors operated by private citizens, small start-up companies or established infrastructure providers can easily be integrated into the platform and the resulting data can be made available to all. The platform – operated by or on behalf of a city – thus forms the basis for an open digitized society, making the city more attractive for its citizens and new businesses, and also helping the city in streamlining and improving its own governmental processes and services.
- **Objective 2: Deploy the City Platform as A Service Solution.** The objective is to develop a Cloud-based open platform for all city information offering multiple services and the capability for federation with other platform instances. Flexibility and elasticity will be key principles in the design of the architecture in order to support the deployment in many different cities with differing requirements, both regarding use cases, services and operational aspects. The platform will also annotate such inherently unreliable IoT data with quality parameters, so that applications can decide if the data quality is good enough to be used.
- **Objective 3: Empower the Citizen to Her Data.** A platform with city information as targeted by CPaaS.io will also have data about its citizens – in particular with the envisioned linking of IoT data with Open Government Data. The platform must thus provide all necessary security mechanisms for the desired privacy and data protection properties to materialize. However, by simply restricting all access to such data, the potential benefits for the individual as well as for the society at large of using such data are lost. CPaaS.io is therefore taking a more progressive stance using a MyData approach: Let the citizen control who can access which personal data under what circumstances.
- **Objective 4: Validate the Platform With Use Cases Providing Public Value.** The platform will be deployed and validated by implementing several use cases that provide public value in cities, be it to the urban society as a whole, be it to individual citizens and city visitors, or be it to the city

² <https://www.cpaas.io/#post-37>

administration. Platform deployments and use case trials will be held in Amsterdam, Sapporo, Yokosuka and Tokyo. While for the validation of the use cases public value and acceptance are the top criteria, the CPaaS.io platform itself will be validated along several additional dimensions: its technical feasibility and performance, ease of deployment and operation, transferability and robustness, and economic benefits for the various stakeholders.

- **Objective 5: Develop Blue Prints for The Adaptation And Transfer Of Solutions To Other Cities.** One of the main reasons for implementing CPaaS.io as a joint EU-Japan project is to demonstrate that concepts and use cases developed in one city can be transferred to other cities even when crossing cultural barriers and different ways in which city, regional and national governments are structured. The project will therefore develop blue prints to make such transfers easy, covering both technical and process aspects as well as the holistic approach to privacy and security across the value chain, independent of the city in which the applications are used and data is exploited.
- **Objective 6: Create Impact in Cities.** Creating impact beyond the partners of the project is in essence the ultimate goal. The project will therefore actively promote the results through its City Stakeholder Group. In addition, public administrations usually place a great importance on systems that conforms to standards, hence the project will emphasize de facto and de jure standardisation in the relevant areas like IoT, Linked Data, Smart City and e-Government.

We have also added an additional column that highlights whether the requirement qualifies as core platform requirement, i.e. a requirement that application stakeholders consider to be fulfilled by the platform and possibly be reused by other applications. Finally, all requirements are prioritized, and requirements used for validation are marked. This supports better tracking throughout the project and upcoming deliverables, e.g. D2.6 Feedback and Validation Report, and D2.7 Criteria for validating federation and transferability. The criteria for selecting which requirements are selected for validation is described in the respective application subsection but is generally influenced by the priority of the requirement and the feasibility of its validation within the project.

In this second version of the requirements specification we have also particularly taken the requirements for cross-border applications into account. These requirements are derived from our work on defining a common scenario that is used to extend the Enhanced User Experience Application (MyEvents web app) with data provided by the Japanese CPaaS.io instance. More specifically, we plan to use data from the U2 architecture that has been collected at the Sapporo Snow Festival 2018. At the time of writing we are negotiating a data transfer agreement between YRP and AGT so that the data can be used and integrated in the MyEvents application.

In addition to the use cases and applications as described in the description of action (DoA) of the Grant Agreement, we also describe the requirements for a Smart Parking application that emerged as a particularly suitable application for transferring project results and experiences to other cities.

The present deliverable is structured as follows: We first describe the Event Management applications including the Enhanced User Experience application in section 2, the Sapporo Visitor Experience in section 3, Tokyo Public Transportation in section 4, and Tokyo Management of Service Vehicles in section 5. In section 6 we provide an update of the requirements of Waterproof Amsterdam and the updated requirements of Yokosuka Emergency Medical Care are described in section 8. Section 9 describes the Smart Parking application including a list of requirements. Finally, we conclude this deliverable with a summary in section 10. Appendix A contains a description of the fields used in the data set description and Appendix B contains a summary of the fields used in the requirements tables.

2 Event Management - Enhanced User Experience

The *Enhanced User Experience application* is one of the applications that are part of the *Managing Fun and Sports Events* Use Case or short *Event Management*. The *Enhanced User Experience* application aims at providing the best possible experiences to event participants. It is a sensor intensive application that draws from wearables and other sensors with the objective to detect event highlights and make event visits for participants unique.

2.1 Application Description

The core idea of this application is to use IoT sensors and analytics to enhance people's experience while visiting or participating at a fun or sports event. Wearables and mobile phones are used as sensors to learn about the activities of event participants. Event participants may include members of the audience, but also performing artists or athletes. For instance, in commercial settings AGT has equipped referees and cheerleaders in basketball matches with wearable sensors and created content based on the analysed data for consumption on site and for distribution via TV broadcasting, social media and other digital distribution channels³. Furthermore, the application uses sensors deployed at the venue to measure and analyse fan behaviour and engagement.

In CPaaS.io we deployed wearables at the Color Run event in Utrecht and created two initial prototypes as described in the prototype release deliverables D2.2 and D2.5. The result is a web-based app, called MyEvents with which users can access highlights of their event participation.

2.2 Updated Scope

While the scope compared to the first version of the requirement specification described in D2.1 has not significantly changed, the updated requirements described in this deliverable are driven by the endeavour of using the application in a joint European Japanese use case and achieving the overall project objectives. In particular, this includes a data collection effort that we conducted at the Sapporo Snow Festival 2018. Details about how we plan to use the MyEvents application as part of the Sapporo Snow Festival and as part of an overall joint use case that is described in deliverable D2.7, which details the criteria for validating federation and transferability.

The scope is driven by the following project objectives:

- **Objective 1: Develop an open social city platform.** The MyEvents application focuses on the social aspects of the platform as it connects personal data collected from individuals both with personal data from friends and aggregated open data. A concrete example for such aggregated open data, are the IoT-based attraction ratings that can be created based on wearables of event participants and made openly available for other applications. In addition to managing personal data, the MyEvents application relies on the CPaaS.io platform to effectively publish new open data.
- **Objective 3: Empower the citizen to her data.** The key data sets for the MyEvents application are personal data. This requires particular care when managing the data in the platform and when transferring the data between regions with different legislations. The experience with the data

³ <http://www.euroleague.net/final-four/berlin-2016/news/i/6vokoibj5fsggg4q/heed-the-event-platform-based-joint-venture-between-wme-img-and-agt-international-makes-first-official-foray-into-sports>

collection efforts in Utrecht and Sapporo further emphasized that effective means of fine granular data access control and the ability to delete all personal data are required.

- **Objective 5: Develop Blue Prints for the adaptation and transfer of solutions to other cities.**

The data collection event at the Sapporo Snow Festival has demonstrated the hurdles of cross-regional data transfer of personal data. Multiple roles of entities collecting and processing the data in different regions using different platform instances create a complexity that requires detailed capturing of how data flows between entities and technical components. In the last year of the project we use the MyEvents application for addressing both, legal and technical challenges associated to managing personal data by different entities.

2.3 Core Functionalities and Components

The implemented architecture has been described in the second prototype release (D2.4) and is repeated in Figure 1 below.

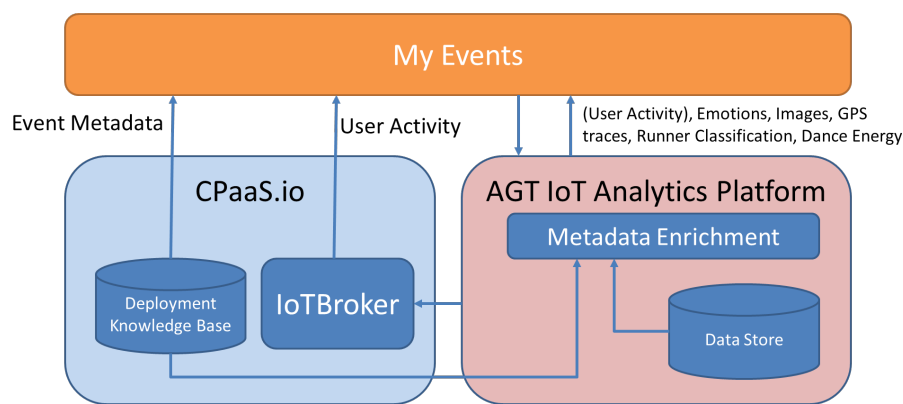


Figure 1: MyEvents Application

Figure 2 depicts the current view on the planned architecture for the MyEvents application. Please note that this reflects the status at the time of writing the deliverable and that the architecture will be subject to the final implementation decisions. The MyEvents application will focus on the semantic integration component that will provide a common interface for accessing both platform instances. Color Run data will be available from the European platform instance while Sapporo Snow Festival data will be available from the Japanese instance.

On the instance of the European platform we plan to connect data sources to AGT's IoT platform that can provide semantically enriched analytic results and provide the data to the IoT Broker. Depending on the configuration of the NGSI to RDF mapper as described in D6.3, the Semantic Integration Layer can use the Semantic Data Repository directly to respond to SPARQL queries from the application or dynamically map between SPARQL queries and NGSI queries and NGSI responses and SPARQL responses.

On the Japanese side we plan to use the SPARQL end point of the U2 architecture. Thus no specific mapping is required. For the MyEvents application the data consumer for the Japanese platform instance is the IoT analytics platform that will provide the processed data to the European instance.

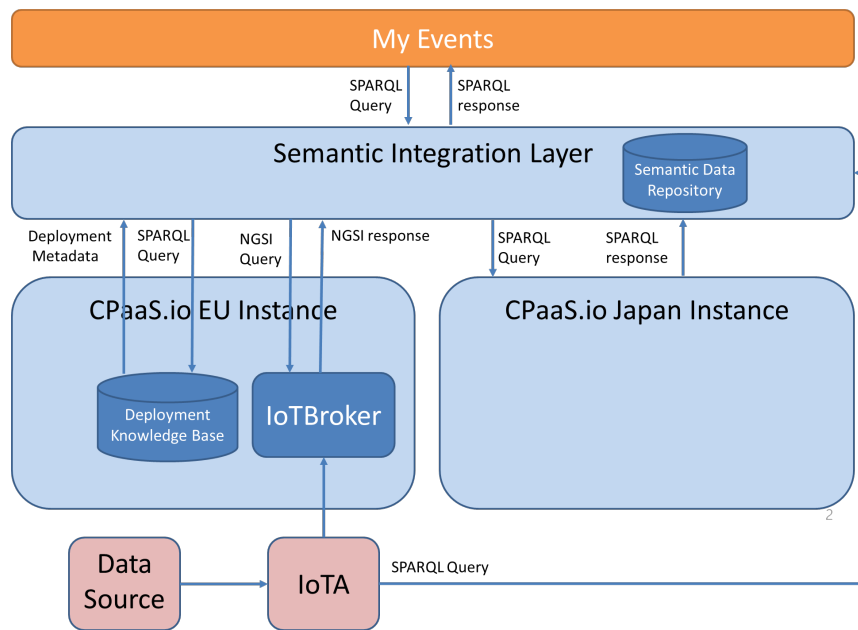


Figure 2: MyEvents architecture reflecting the updated scope

Please note that the following main dependencies and requirements need to be taken into consideration when moving towards the envisaged architecture as described above:

1. **The security components of WP5 need to be integrated with the IoTBroker.** Without encrypting the data managed by the IoT broker and providing fine grained access control, we cannot provide personal data to CPaaS.io.
2. **The data transfer agreement between YRP and AGT needs to be signed.** Without the signature of the data transfer agreement, we cannot transfer the data between the Japanese and European platform instance.
3. **The semantic integration layer needs to be able to query the appropriate platform instance.** The semantic integration layer needs to have knowledge about which data store holds the requested information in order to respond with the correct results.

2.4 Data Sets

Although the scope of the MyEvents application has been extended to cover the Sapporo Snow Festival, there are no major changes with respect to the sensor-based data sets used by the application. The same sensors and data sets have been used for collecting the data.

2.5 Deployment Plan

The deployment plan is aligned with the initial deployment plan as described in D2.1 which we repeat here for convenience:

- M01-M03: Initial requirement analysis and 1st deployment in order to collect data (D2.1)
- M03-M06: Data analysis and implementation of first analytics prototype (D2.2)
- M06-M12: 2nd deployment including test of analytics chain (internal milestone)

The first deployment was executed at the Color Run Utrecht 2016, the analytics chain has been implemented and analytics results been made available as described in the second prototype release. The second deployment has been executed at the Sapporo Snow Festival 2018. As soon as the data transfer

agreement is in place, we will investigate the collected data and explore whether new features can be integrated into the application.

2.6 Requirements

In this section we extend on the requirement described in the first version (D2.1). The list of requirements has not changed, but we extend on the existing requirements by adding new fields as described in Appendix B. In addition, we provide some additional explanatory text for each of the requirements.

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for validation
EUE-1	Data collection infrastructure	All services require the collection of real-time data	Existence of a data collection infrastructure	Old	Yes	High	O1	Done	Yes
EUE-2	Metadata management	Sensor data can only be processed if it is known how sensors were deployed, operated, to which user, event etc. the data relates etc.	Method for collecting metadata and enriching sensor data	Old	Yes	High	O1	Done	Yes
EUE-3	Visualization of personal data	The visualization component is the front end towards end user that provides him/her access to a record of his/her event experience.	Visual data access for end user	Old	No	High	O1, O3	Done	No
EUE-4	Data processing	Data such as excitement levels require processing of raw sensor data	Availability of analytical services	Old	Yes	High	O1, O3	Done	Yes
EUE-5	Control over data collection	This allows users to be empowered to make a decision between what data is collected and what services they can use	UI for controlling use of data	Old	Yes	Medium	O3	Open	Yes

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for validation
EUE-6	Synchronized clocks between IoT devices	Time synchronization is required so that sensor data from different sensors can be correlated by time	Maximum clock drift	Old	No	High	O4	Done	No
EUE-7	Efficient query of user data	Personalized visualizations require efficient retrieval of media data and processed data from sensor streams	Query latency in ms	Old	Yes	High	O1	Partially done	Yes
EUE-8	Guarantees for base functionality	When local (BT links, wireless links to local base stations) or backend connectivity (cloud) is lost, the application must still provide value to the end user	User value	Old	No	Low	O4	Done	No
EUE-9	Data completeness	When local (BT links, wireless links to local base stations) or backend connectivity (cloud) is lost, it must be ensured that data loss is avoided in order to ensure service operation	Percentage of lost packets	Old	No	High	O4	Done	No

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for validation
EUE-10	Data sharing	Users may want to share their data via various channels using various platform instances	Availability of data sharing mechanisms	Old	Yes	Low	O3	Open	No
EUE-11	Semantic data integration	Data sharing between diverse systems can only efficiently be achieved if the semantics of the data are machine readable	Ontologies for data to be shared	Old	Yes	High	O5	Partially Done	Yes
EUE-12	Scale to thousands of people	The application will collect sensor and media data for thousands of people. The platform needs to be able to efficiently ingest, store, process user data.	Number of data points ingested per second, processing latency, storage footprint	Updated	Yes	Medium	O6	Started	No
EUE-13	Access data from different platform instances	Depending on where event takes place and who is providing the data, it may be stored in different platform instances. Ideally this should be transparent for the application.	Transparent data access across platform instances	New	Yes	High	O6	Started	Yes

2.6.1 Requirements Used for Validation

The following requirements are used for validating the platform. They have been selected for validation as 1) they have been identified as a core requirement that should be provided by the platform, and 2) due to their high priority.

2.6.1.1 EUE-1

The data collection infrastructure is currently realized as part of the proprietary data collection infrastructure of AGT's IoT platform.

2.6.1.2 EUE-2

The metadata is managed and accessed by the CPaaS.io deployment knowledge base.

2.6.1.3 EUE-4

The data processing component has been realized as an analytical component in AGT's IoT analytics platform. Consequently, there is not a strong requirement towards the platform to provide a data processing or analytics service. This requirement links to objective 1, because the question of how to integrate analytic results is an important challenge that CPaaS.io needs to address for developing the overall platform. It also links to objective 3, as users must be able to control both their raw and processed data.

2.6.1.4 EUE-5

This requirement is extremely important for managing personal data. Our experiences in the Utrecht and Sapporo data collection effort have shown that the current approach of manual consent forms, data transfer agreements, and implementing the technical and organizational measures slows down the process of using the data considerably. The whole process of managing personal data is extremely complex due to the involvement of multiple data controllers, data processors and data subjects spread across regions with different legislation. It is therefore highly desirable that the platform facilitates this process.

2.6.1.5 EUE-7

In the current implementation, the query latency based on the IoT platform fully satisfies the application requirements. We have also conducted some initial measurements on the IoTBroker that seems to satisfy the requirement at least for smaller data sets. The maximum query latency should be below 1s.

2.6.1.6 EUE-11

As described in D6.3, ontologies for describing the deployment aspects of sensors have been defined. In addition, in section 2.3 we have described how the SPARQL and the semantic integration layer is used by the application to share data between platform instances. A basic ontology for sharing MyEvent data is currently under development.

2.6.1.7 EUE-13

This requirement is a new requirement that emerged from the fact that the Enhanced User Experience application will need to access both the European and Japanese platform instance in order to cover the Sapporo Snow Festival event. Transparent semantic data access to federated platform instances is a core requirement to the CPaaS.io platform.

2.6.2 Requirements Not Used For Validation

The following list of requirements are not used for validation. For each requirement we describe briefly why.

2.6.2.1 EUE-3

The visualization component has been realized as part of the second prototype release D2.5. This requirement is not used for validating the platform as it requires the implementation of an application specific component.

2.6.2.2 EUE-6

This requirement has been implemented by using the Network Time Protocol (NTP) and is implemented by the data sources and the IoT platform. Thus, it is not a requirement towards the platform and therefore not used for validation.

2.6.2.3 EUE-8

The MyEvents application is a web-based application and the requirement is satisfied by caching the data in the web browser. This requirement is not used for validation as it is not a core platform requirement.

2.6.2.4 EUE-9

Data completeness is achieved by the data collection infrastructure of AGT's analytics platform, which includes a mobile app that buffers data locally if there is no network connectivity. As this is not a core platform requirement it is not used for platform validation.

2.6.2.5 EUE-10

For the application itself it is not necessary that users can share their data. However, it would empower users to better control whether friends can use for instance a dance energy index provided by the platform to compare against their own instance. In addition, such a functionality would help in digitalizing and implementing informed user consent. Due to the low priority of the requirement it is not used for validation.

2.6.2.6 EUE-12

This requirement has been updated to scale with data collected from thousands of people at the same time. For instance, if we assume that about 10.000 people participate at a colour run event and 10% of the participants are equipped with sensors, we need to target collecting and processing the data from about 1.000 participants at the same time. This requirement has not been selected for validation due to its medium priority.

3 Event Management – Sapporo Visitor Experience

The Sapporo *Visitor Experience application* is part of a set of applications that will use the CPaaS.io platform to manage fun and sports event. This application focuses on tourist information services including event information and public transportation information.

3.1 Application Description

See V1 (D2.1) document since there have not been any changes.

3.2 Architecture

See V1 (D2.1) document since there have not been any changes.

3.3 User Interface

See V1 (D2.1) document since there have not been any changes.

3.4 Data Sets

Although there have not been changes for the data sets since V1 (D2.1), for convenience we quote the paragraphs verbatim.

We established the “Sapporo Open Data Association”, whose members are data providers such as Sapporo City, companies and organizations in tourism, sports and transportations. The association collects data from its members, transforms them to Open Data and publishes them on the Sapporo Open Data Portal.

Sightseeing Spot	
Detailed Description	Comments, pictures and locations of popular sightseeing spots
OGD or private data	OGD
Personal Data	No
Hosting	Sapporo Open Data Association
Data Provider	Sapporo City
Format	PDF, HTML, CSV, JPEG
Update Frequency	Yearly
Update Size	N/A
Data Source	Web
Sensor	unicode marker
Number of Sensors	11

Meal	
Detailed Description	Restaurant name, location, opening hours, menu, service
OGD or private data	Private
Personal Data	No
Hosting	Sapporo Open Data Association
Data Provider	Hotel, Sapporo Station Shopping Mall
Format	PDF, HTML, CSV, JPEG
Update Frequency	Monthly
Update Size	N/A
Data Source	Web
Sensor	unicode marker
Number of Sensors	11

Hotel	
Detailed Description	Hotel name, location, number of rooms, access
OGD or private data	Private
Personal Data	No
Hosting	Sapporo Open Data Association
Data Provider	Hotels
Format	PDF, HTML, CSV, JPEG
Update Frequency	Monthly
Update Size	N/A
Data Source	Paper
Sensor	N/A
Number of Sensors	N/A

Other Facilities	
Detailed Description	Location of ATM, Toilet, Rental Locker, etc.
OGD or private data	Private
Personal Data	No
Hosting	Sapporo Open Data Association
Data Provider	Sapporo Station Shopping Mall
Format	PDF, HTML, CSV, JPEG
Update Frequency	Monthly
Update Size	N/A
Data Source	Web
Sensor	unicode marker
Number of Sensors	11

Snow Festival	
Detailed Description	Event information in Sapporo Snow Festival
OGD or private data	Private
Personal Data	No
Hosting	Sapporo Open Data Association
Data Provider	Operating Committee of Sapporo Snow Festival
Format	PDF, HTML, CSV, JPEG
Update Frequency	Yearly
Update Size	
Data Source	Web
Sensor	unicode marker
Number of Sensors	11

Stadium Data	
Detailed Description	Location, facilities and drawing of stadiums
OGD or private data	OGD
Personal Data	No
Hosting	Sapporo Open Data Association
Data Provider	Sapporo City
Format	PDF, HTML, CSV, JPEG
Update Frequency	Yearly
Update Size	N/A
Data Source	Web, Paper
Sensor	N/A
Number of Sensors	N/A

Train & Bus Data	
Detailed Description	Time table, Operating status, Line, Real time location, bus stop, etc.
OGD or private data	OGD & private
Personal Data	No
Hosting	Sapporo Open Data Association
Data Provider	Sapporo City, Hokkaido Chuo Bus
Format	PDF, HTML, CSV, JPEG
Update Frequency	Monthly
Update Size	N/A
Data Source	File

Sensor	N/A
Number of Sensors	N/A

Express Way Data	
Detailed Description	Information of service area
OGD or private data	Private
Personal Data	No
Hosting	Sapporo Open Data Association
Data Provider	NEXCO East Japan
Format	PDF, HTML, CSV, JPEG
Update Frequency	Yearly
Update Size	N/A
Data Source	Web
Sensor	N/A
Number of Sensors	N/A

Underground Map Data	
Detailed Description	Drawing of underground map
OGD or private data	Private
Personal Data	No
Hosting	Sapporo Open Data Association
Data Provider	Sapporo Station Shopping Mall
Format	PDF, HTML, CSV, JPEG
Update Frequency	Yearly
Update Size	N/A
Data Source	File
Sensor	unicode marker
Number of Sensors	11

3.5 Deployment Plan

We have started information services on Kokosil before Sapporo Snow Festival and Asian Winter Games in February 2017. At the event venue, we will introduce the Kokosil service to visitors and collect feedback from users. Additionally, data for the Enhanced User Experience application was collected at the Sapporo Snow Festival in February 2018 as explained in section 2.

Also, we plan to carry out additional prototyping using signages in public places in Sapporo pending the cooperation of the city of Sapporo who owns and manages the public spaces.

3.6 Requirements

Table 1 lists the requirements.

Table 1: Requirements for Sapporo Visitor Experience

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
SVE-1	Data collection infrastructure	The service requires the collection of real-time data about the visitor (for example, the current position, preferred language for guidance, etc.) We include "Communication " as part of this infrastructure	Existence of a data collection infrastructure	New	Yes	Low	O1	Done	Yes
SVE-2	Metadata management	Sensor data and data from open data sources of various players such as transportation operators and government tourism promotion office can only be processed if it is known how sensors were deployed, operated, to which user, event etc. the data relates etc.	Method for collecting metadata and enriching sensor data	New	Yes	High	O1	Done	Yes
SVE-3	Visualization of personal data and open data	The visualization component is the front end towards end user (tourists, tourism business operators, etc.) that provides them access to a record of the tourist, and sightseeing and tourism information.	Visual data access for end user	New	No	O1, O2	O1, O3	Done.	Yes
SVE-4	Data processing	Sapporo visitor experience as originally envisioned doesn't need this much	Availability of analytical services	New	Yes	low	O1, O2	Done	No

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
SVE-5	Control over data collection	This allows tourists to be empowered to make a decision between what data is collected while the tourists are in the service area.	UI for controlling use of data Aside from automatically collected data such as GPS location, the tourists can enter data such as places in which they are interested, their food preference, etc. at their discretion.	New	Yes	Medium	O3	Done partially	No
SVE-6	Synchronized clocks between IoT devices	Time synchronization is required so that sensor data from different sensors can be correlated by time	Maximum clock drift (usually second resolution is sufficient for tourism purposes)	New	No	High	O1	Done	No
SVE-7	Efficient query of sightseeing information	Personalized visualizations require efficient retrieval of media data and processed data from sensor streams. (For example, map information needs to be shown quickly. Also, some guidance video clips are often shown.)	Query latency in sec.	New	Yes	High	O1	done	Yes
SVE-8	Guarantees for base functionality	When communication between the cloud servers and mobile terminals of tourists are cut off, the cached information can be used until the communication is restored.	User value	New	No	Low	O1	Done.	No
SVE-9	Data completeness	SVE as originally envisioned does not use many sensors on the tourists' side and so this may not apply here.	Percentage of lost packets	New	No	Low or N/A	O1	Done.	No
SVE-10	Data sharing	Tourists may want their preference such as place of interest, their food preference, preferred language for guidance shared between service operators in the region.	Availability of data sharing mechanisms	New	Yes	Medium	O3	Done partially.	No

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
SVE-11	Semantic data integration	The data offered by various tourism service operators need to be offered uniformly and this requires the data integration function. But this requires cooperation from the tourism business operators who provides open data.	Ontologies for data to be shared	New	Yes	High	O4	Open	No
SVE-12	Scale to a few million visitors in one week	The largest event in Sapporo, Sapporo Snow Festival attracts a few million people in a week. So the system needs to support such large user base.	Number of tourists profile registered over time, processing latency, storage footprint of such tourist data. And the same goes for the sightseeing information provided by the service operators and accessed by such tourists via kokosil.	New	Yes	Medium	O6	Done.	No

3.6.1 Requirements Used for Validation

The following requirements are used for validating the platform. They have been selected for validation as 1) they have been identified as a core requirement that should be provided by the platform, and 2) due to their high priority. Some core requirements could not be validated due to scalability issues.

3.6.1.1 SVE-1 Data collection infrastructure

The data collection infrastructure is currently realized as described in the high-level application architecture (see D2.1) and marker information is also collected to learn the whereabouts of users in underground passages.

3.6.1.2 SVE-2 Metadata management

The metadata is managed and accessed by the high-level application architecture (see D2.1).

3.6.1.3 SVE-3 Visualization of personal data and open data

Open data related to tourism and public transportation is incorporated and used for user interaction. (See D2.1). Personal data is not incorporated yet. (being planned for the future).

3.6.1.4 SVE-7 Efficient query of Sightseeing information

In the current implementation, the query latency based on the high-level application architecture fully satisfies the visitor experience application requirements. This requirement is extremely important for providing tourism information as envisioned by the original application.

3.6.2 Requirements not used for validation

The following list of requirements are not used for validation. For each requirement we describe briefly why.

3.6.2.1 SVE-4 Data processing

Data processing is not needed much in Sapporo User Experience prototype.

3.6.2.2 SVE-5 Control over data collection

The current application does not take advantage of personal data very much and thus this criteria was not used for validation. Incorporation of u2 implementation of CPaaS.io PDS is being planned, and once that happens, we can offer more customized service to the users, but since the basic functions of high-level application are already available, this is not used for validation.

3.6.2.3 SVE-6 Synchronized clocks between IoT devices

Synchronization is important generally, however, as of now, the application does not need sub-second accuracy of clocks. So this is not used for validation at this stage.

3.6.2.4 SVE-8 Guarantees for base functionality

Guarantees for base functionality in the face of loss of communication, cached information can be used to some extent. But this is not used for validation purposes as of now.

3.6.2.5 SVE-9 Data completeness

SVE as originally envisioned does not use many sensors on the tourists' side and so this is not a strong validation factor.

3.6.2.6 SVE-10 Data Sharing

Tourists may want their preference such as place of interest, their food preference, preferred language for guidance shared between service operators in the region. As long as the service is offered through the common platform kokosil, the data is shared. Thus it is done partially. However, as is clear, this requires the wide cooperation of operators in the region, and is not considered for validation factor at this stage.

3.6.2.7 SVE-11 Semantic data integration

The data offered by various tourism service operators need to be offered uniformly and this requires the data integration function. But this requires cooperation from the tourism business operators who provides open data. The cooperation is beyond CPaaS.io project control and so this was not used for validation.

3.6.2.8 SVE-12 Scale to a few million visitors in one week

In principle, there does not seem to be inherent scalability issue. But CPaaS.io did not have the resources to try this out for real. It is up to the local city and tourism industry to use the high-level application architecture that includes kokosil as a component at the forecasted scale. This this was not used for validation.

4 Event Management – Tokyo Public Transportation

The *Tokyo Public Transport application* is part of a set of applications that will use the CPaaS.io platform to manage fun and sports event. This sub-application focuses on integrating data from different public transportation services.

4.1 Application Description

See V1 (D2.1) since there have not been changes.

This use case trial shall be performed in Tokyo Area with the support of the "Association of Open Data for Public Transportation" (AODPT⁴). As of February 2018, ODPT has 60 members. The number of members has increased almost two-fold since the CPaaS.io proposal had been filed. The members include railway companies, bus companies, Japan Airlines, All Nippon Airways, and ICT companies such as Fujitsu, NEC, Google, and Microsoft. We are collaborating with this association for the deployment of the results of the CPaaS.io project.

4.2 Architecture

See V1 (D2.1) since there have not been any changes.

4.3 User Interface

See V1(D2.1) since there have not been any changes. Please note there will be many different UIs for different apps (for example, ones developed for "Open Data Challenge for Public Transportation in Tokyo" contest (see 4.5 for details).

4.4 Data Sets

The data sets have not changed since V1 (D2.1), but for convenience, we list them here.

In addition to the collaboration with ODPT, we will also have support from the Ministry of Land, Infrastructure, Transport and Tourism which releases many kinds of open data for public transportation. In our application, we will use data sets as summarized in the following table:

⁴ <http://odpt.org>

Tokyo Public Transportation Data	
Detailed Description	(1) Time tables and real-time traffic information of public transportation such as railways, buses, and airplanes. (2) Geographical data for routes, locations of railway stations, bus stops, airports, etc. (3) Data of railway stations such as maps, facility lists,
OGD or private data	OGD and Private Data
Personal Data	Not included
Hosting	Consortium of Open Data for Public Transportation
Data Provider	Consortium of Open Data for Public Transportation, and many public transportation companies and sectors
Format	ucR (ucode Relation) format (RDF + ucode in JSON format)
Update Frequency	Yearly: Geographical data, railway station data, etc. About every 6 months: Static data for operation such as time tables, etc. Real-time update: real-time traffic information, etc.
Update Size	New real-time data is appended to (old) existing data. Yearly update size and 6-month update size are estimated to be relatively small (less than 10% of the existing data size).
Data Source	1. Tokyo Public Transportation Open Data http://www.odpt.org/ 2. Tokyo Metro Open Data https://developer.tokyometroapp.jp/info 3. Open Data for Pedestrian Navigation Support Services https://www.hokoukukan.go.jp/top.html 4. 4. others
Sensor	Location sensors of trains, GPS for bus locations, etc.
Number of Sensors	Hard to count/ hard to estimate (To output the data, all train and bus cars are using sensors.)

4.5 Deployment Plan

As the first step, we have started to offer public transportation open data services in Tokyo in collaboration with ODPT. The final target year is 2020, in which the Tokyo Olympic and Paralympic Games will be held in Tokyo. For the next step, we will provide SaaS for public transportation for many satellite cities in Japan.

In order to test the feasibility of the system approach, we have organized a contest for the development of applications based on open data made available by ODPT, called the "Open Data Challenge for Public Transportation in Tokyo".⁵

This contest will help us to iron out the bugs and undesirable features from the open data offering. Also, the apps developed in the contest will prove (or disprove) the realistic applicability of our approach.

This public transportation application scenario is planned to be experimented in the third year of the CPaaS.io project.

⁵ See <https://tokyochallenge.odpt.org/en/index.html> for details.

4.6 Requirements

The table below lists the updated requirements. In addition to the original requirements listed in V1 (D2.1), we have looked at the requirements listed for EU use cases and added appropriate relevant items to the list.

ID	Description	Rationale	Fit Criterion	Old/New /Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
TPT-1	Automated data collection mechanism	Public transportation data tends to update frequently. Efficient update of the data requires an automatic data collection system (even for static data such as time tables and maps of stations).	Existence of automatic data collection system for public transportation.	Old	Yes	High	O1	Done	No
TPT-2	Multi-lingual data	Foreign tourists may not have Japanese language skills and need multilingual information on public transportation.	Ratio of multilingual information in all information. Existence of automatic translation system.	Old	Yes	High	O6	Done partially	Yes
TPT-3	Live data collection and distribution	This use case deals with real-time operation information of public transportation such as trains, busses, and air planes. It is necessary to provide automatic data collection mechanism for these data.	Existence of automatic collection system and distribution system of live operation data.	Old	Yes	High	O1	Done	Yes
TPT-4	Data format and API standardization (Metadata + API)	Since public transportation is operated by many private and public sectors, data distribution can only be achieved efficiently based on standard data formats and APIs. Sensor data and open data from public transportation operators can only be processed if it is known how sensors were deployed, operated, to which user, event etc. the data relates etc. End user information needs such metadata, too.	Existence of standard data format and standard API specifications.	Updated	Yes	High	O1,O5	Done partially	Yes (for compliant operators)

ID	Description	Rationale	Fit Criterion	Old/New /Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
TPT-5	Secure and robust communication	Public transportation is a critical social infrastructure. Communication among players of the public transportation open data system should be secure and robust.	Existence of cryptographic communication mechanism and fault tolerance mechanism.	Old	Yes	High	O6		Yes
TPT-6	Developers' Site	For the open innovation with public transportation open data, it is necessary to provide an open development environment on the platform called "developers site".	Existence of developers' site of Tokyo Public Transportation Open Data.	Old	Yes	High	O1,O6		Yes
TPT-7	Data License	Good data license is necessary for the deployment of the data. For the data license, the balance between data holders' merits and users' merits is essentially important.	Existence of open data license.	Old	Yes	High	O1		Yes
TPT-8	Third party apps	Open data is a method for the collaboration between data holders and application developers.	Number of third party apps or software.	Old	Yes	High	O1, O6		Yes
TPT-9	Services for the physically challenged	Contribution to improve quality of life of the physically challenged and the elder in moving in Tokyo is one of the most important objectives.	Number of services for the physically challenged.	Old	Yes	High	O4		No
TPT-10	User feedback collection	User feedback is necessary for the evaluation of the use case experiment.	Number of user feedbacks.	Old	Yes	High	O6		No

ID	Description	Rationale	Fit Criterion	Old/New /Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
TPT-11	Visualization of data	The visualization component is the front end for end user that provides them access to interesting data such as timetable, the location of vehicles, operation status, etc.	Visual data access for end user	New	No	High	O3	Done partially	No
TPT-12	Data processing	The open data from public transportation operators need to be processed to produce interesting output to the end users (for example, finding the shortest path between location A and B, or the fastest service between two locations, etc.)	Availability of analytical services	New	Yes	medium	O1, O3	Done partially	No
TPT-13	Control over data collection	This allows users to be empowered to make a decision about what data is collected while the end user is in Tokyo region (such as automatically collected GPS location, the destination of a trip, etc.)	UI for controlling use of data	New	Yes	High	O3	Done partially	No
TPT-14	Synchronized clocks between IoT devices	Time synchronization is required so that sensor data from different sensors can be correlated by time	Maximum clock drift (usually milliseconds.)	New	No	High	O4	Done.	No

ID	Description	Rationale	Fit Criterion	Old/New /Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
TPT-15	Efficient query of transportation open data	Timely visualization of interesting data requires efficient retrieval of sensor and open data and processed data from sensor streams and data sources.	Query latency in sec.	New	Yes	High	O1	Done (for currently available open data sources: not all public transportation operators have released their data yet.)	Yes
TPT-16	Guarantees for base functionality	When communication between the cloud and the mobile terminal of end users is cut off, cached data can be used to some extent. But the nature of transportation information which requires the always up-to-date status of the transportation service, the loss of communication basically means that the service is degraded very badly.	User value	New	No	Low	O4	Done.	No
TPT-17	Data completeness	This actually falls within the responsibility of public transportation operators who provide the data to end users.	Percentage of lost packets	New	No	High	O4	Descoped	No
TPT-18	Data sharing	Data from various data sources need to be shared. End user information needs to be shared with certain limit.	Availability of data sharing mechanisms	New	Yes	High	O1, O3	Done	Yes

ID	Description	Rationale	Fit Criterion	Old/New /Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
TPT-19	Semantic data integration	The data from various data sources must be integrated and offered in a uniform manner.	Ontologies for data to be shared	New	Yes	High	O5	Done	Yes
TPT-20	Scale to millions of users	The data sources are expected to be used by a few million users daily. Each day, about 3 million people pass through Tokyo's Shinjuku station. So the system as a whole need to support such scalability.	Number of data points ingested per second, processing latency, storage footprint, and the number of queries	New	Yes	Medium	O6	Descoped	No
TPT-21	Communication	If the communication between the cloud servers and the mobile terminal of end users is cut off, then cached information may be usable to some extent. However, due to the nature of transportation data which needs always up-to-date status, such cached data may not be useful all the time.	Number of data queries per seconds, processing latency storage footprint, and required	New	Yes	Medium	O4, O6	Done	No

4.6.1 Requirements Used for Validation

The following requirements are used for validating the platform. They have been selected for validation as 1) they have been identified as a core requirement that should be provided by the platform, and 2) due to their high priority. Some core requirements could not be validated due to scalability issues and also because those requirements must be satisfied by the public transportation operators themselves and not by the CPaaS.io partners.

4.6.1.1 TPT-2 Multi-lingual data

Data processing is not needed much in Sapporo User Experience prototype.

4.6.1.2 TPT-3 Live Data collection and distribution

This use case deals with real-time operation information of public transportation such as trains, busses, and air planes. It is necessary to provide automatic data collection mechanism for these data and provide the data to the users.

4.6.1.3 TPT-4 Data format and API standardization (Metadata + API)

Done partially. Not every ODTP public transportation operators have complied yet. This is beyond the control of CPaaS.io partners. Since public transportation is operated by many private and public sectors, data distribution can only be achieved efficiently based on standard data formats and APIs. Sensor data and open data from public transportation operators can only be processed if it is known how sensors were deployed, operated, to which user, event etc. the data relates etc. End user information needs such metadata, too. This is very important for data distribution.

4.6.1.4 TPT-5 Secure and robust communication

Public transportation is a critical social infrastructure. Communication among players of the public transportation open data system should be secure and robust.

4.6.1.5 TPT-6 Developers site

For the open innovation with public transportation open data, it is necessary to provide an open development environment on the platform called "developers site".

4.6.1.6 TPT-7 Open data license

Good data license is necessary for the deployment of the data. For the data license, the balance between data holders' merits and users' merits is essential.

4.6.1.7 TPT-8 Third party apps

Open data is a method for the collaboration between data holders and application developers. So the more number of apps means that the open data infrastructure is more successful.

4.6.1.8 TPT-15 Efficient query of transportation open data

Done for currently available data sources.: not all public transportation operators have released their data yet. Timely visualization of interesting data requires efficient retrieval of sensor and open data and processed data from sensor streams and data sources.

4.6.1.9 TPT-18 Data sharing

Done for the currently available open data sources. Data from various data sources need to be shared. End user information needs to be shared to some extent (but not much interest as of now from the viewpoint of CPaaS.io project).

4.6.1.10 TPT-19 Semantic data integration

Done for the currently available data sources. The data from various data sources must be integrated and offered in a uniform manner. Note that this is primarily the responsibility of the public transportation operators.

4.6.2 Requirements NOT Used for Validation

The following list of requirements are not used for validation. For each requirement we describe briefly why.

4.6.2.1 TPT-1 Automated data collection mechanism

Automated data collection mechanism. This refers to the data collection done by the transportation operators. Although it is very important, this is beyond the control of CPaaS.io partners in Japan. Thus it is excluded from the validation criteria.

4.6.2.2 TPT-9 Services for the physically challenged

This is important for future cities. However, the provision of the services depend on the data providers, i.e., transportation operators: for example, we depend on them to provide the barrier-free routes inside the station for wheelchair users. Such provision is beyond the control of CPaaS.io partners. Thus it is excluded from the validation criteria. It is certain that we can offer such services once the data for them is provided by the public transportation operators.

4.6.2.3 TPT-10 User feedback

At this stage, the public transportation operators have just begun to accept the open data approach. They are not quite ready to listen to the user feedback based on the use of the published open data. So the implementation of user feedback loop either via software or manual survey is simple in principle, we have excluded this from validation criteria.

4.6.2.4 TPT-11 Visualization of data

Done for certain applications. The visualization component is the front end for end user that provides them access to interesting data such as timetable, the location of vehicles, operation status, etc. However, this is more about the APPLICATION built on top of CPaaS.io infrastructure and not about the underlying CPaaS.io infrastructure itself. Thus, we have excluded this from the validation criteria.

4.6.2.5 TPT-12 Data processing

Done for certain applications. The open data from public transportation operators need to be processed to produce interesting output to the end users (for example, finding the shortest path between location A and B, or the fastest service between two locations, etc.) However, this is more about the APPLICATION built on top of CPaaS.io infrastructure and not about the underlying CPaaS.io infrastructure itself. Thus, we have excluded this from the validation criteria.

4.6.2.6 TPT-13 Control over data collection

Done for certain applications. This allows users to be empowered to make a decision about what data is collected while the end user is in Tokyo region (such as automatically collected GPS location, the destination of a trip, etc.) However, this is more about the APPLICATION built on top of CPaaS.io infrastructure and not about the underlying CPaaS.io infrastructure itself. Thus, we have excluded this from the validation criteria.

4.6.2.7 TPT-14 Synchronized clocks between IoT devices

Time synchronization is required so that sensor data from different sensors can be correlated by time. However, this is to be done by public transportation operators, and NOT BY CPaaS.io consortium. Thus, we have excluded this from the validation criteria.

4.6.2.8 TPT-16 Guarantees for base functionality

When communication between the cloud and the mobile terminal of end users is cut off, cached data can be used to some extent. But the nature of transportation information which requires the always up-to-date status of the transportation service, the loss of communication basically means that the service is degraded very badly. Since there is not much we can do in the face of communication cut-off, we excluded this from the validation criteria.

4.6.2.9 TPT-17 Data completeness

This actually falls within the responsibility of public transportation operators who provide the data to end users. This is beyond the responsibility of CPaaS.io partners. Thus, we have excluded this from the validation criteria.

4.6.2.10 TPT-20 Scale to millions of users

Descoped. The data sources are expected to be used by a few million users daily. Each day, about 3 million people pass through Tokyo's Shinjuku station. So the system as a whole need to support such scalability. A single service app may not need to serve million users: an app may only have a few hundred thousand users, but there can be many service apps that access the data sources simultaneously. While the architecture should be capable of supporting such capability, Due to resource issues, testing at such a scale won't be done within CPaaS.io context. Thus, we have excluded this from the validation criteria.

4.6.2.11 TPT-21 Communication

If the communication between the cloud servers and the mobile terminal of end users is cut off, then cached information may be usable to some extent. However, due to the nature of transportation data which needs always up-to-date status, such cached data may not be useful all the time. Since there is not much we can do in the face of communication cut-off, we excluded this from the validation criteria.

5 Event Management – Tokyo Management of Service Vehicle

The *Tokyo Public Transport application* is part of a set of applications that will use the CPaaS.io platform to manage city events. This sub-application focuses on ensuring reliable transportation of a wide range of vehicles in the city including for instance utility vehicles by private companies.

5.1 Application Description

This use case trial shall be performed in the extended Tokyo Area with a private transportation company. It utilizes the package gathering/delivery operation in a metropolitan area. It provides plan, operation, and monitor of service vehicles and images of truck beds. It provides real time updates of delivery plan and monitoring of vehicle positions on a map. It provides QR-code based driver registration on a smart phone application at the driver side.

5.2 Architecture

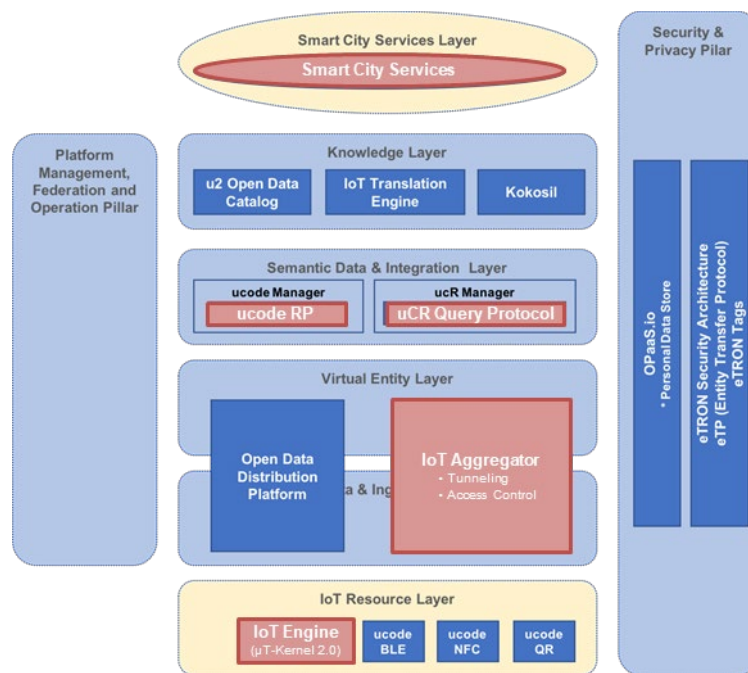


Figure. u2-based architecture

5.3 User Interface

The user interface will be provided as a) a mobile web application and b) a web browser application.

A smart phone UI for that provides driver assistance of vehicle transportation. There will also be an integrated web application to incorporate the real-time operation, pre-defined routes and plans, real-time plan updates, and road information such as traffic conditions and weather forecast.

A web browser application provides plan operator assistance of delivery planning, update, and real time monitoring of delivery updates.

5.4 Data Sets

There are four types of data:

- Operation plan
- Operation logs

Operation plan	
Detailed Description	Routes, delivery time and place, contact addresses, driver information
OGD or private data	Private
Personal Data	Contact addresses, driver information
Hosting	External
Data Provider	Operation companies
Format	JSON
Update Frequency	Daily (with some irregular updates)
Update Size	KB

Operation plan	
Data Source	Operation companies
Sensor	N/A
Number of Sensors	N/A

Vehicle logs	
Detailed Description	Tracking data of vehicle transportation
OGD or private data	Private
Personal Data	No
Hosting	External
Data Provider	Driver of a vehicle
Format	JSON
Update Frequency	Per minute
Update Size	KB
Data Source	Smartphone of a driver
Sensor	Embedded in a smartphone
Number of Sensors	1 per smartphone

Truck bed status	
Detailed Description	Camera Photo of Truck beds
OGD or private data	Private
Personal Data	No
Hosting	External
Data Provider	Service providers
Format	image
Update Frequency	Per gathering/delivery (1 per 1 hour, approx..)
Update Size	50 KB
Data Source	Service providers
Sensor	Camera embedded in a smartphone
Number of Sensors	Not known

5.5 Deployment Plan

The deployment is planned for the year 3. Detailed adoption plan is under way at the time of this writing.

In the simplest case, vehicle location is tracked by the smartphone of the driver.

ACCESS has experience of implementing vehicle transportation planning and management systems.

In order to provide enhanced user experience, iterated cycles of field experiments and integration of city information data has been deployed using the CPaaS.io platform.

The deployment and improvement cycles of field tests will provide an important step-stone to provide robust and reliable operation-management services on top of a city information infrastructure platform. It will test the feasibility of a city information infrastructure for a wide range of operation services deployed in city areas.

5.6 Requirements

Table 2 lists the requirements for managing service vehicles in Tokyo.

Table 2: Requirements for Tokyo Management of Service Vehicles

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
TMS-1	Automated data collection mechanism	Transportation data tends to update frequently. Efficient update of the data requires an automatic data collection system (even for static data such as time tables and maps of stations).	Existence of automatic data collection system for public transportation.	Old	Yes	High	O1	Open	No
TMS-2	Multi-lingual data	Foreign labour may not have Japanese language skills and need multilingual information on transportation.	Ratio of multilingual information in all information. Existence of automatic translation system.	Old	Yes	Low	O6	Described	No
TMS-3	Live data collection and distribution	This use case deals with real-time operation information of transportation between a driver and a operator.	Existence of automatic collection system and distribution system of live operation data.	Old	Yes	High	O1	Done	Yes
TMS-4	Data format and API standardization (Metadata + API)	Since transportation is operated by many private companies, data distribution can only be achieved efficiently based on standard data formats and APIs. Sensor data and open data from public transportation operators can only be processed if it is known how sensors were deployed, operated, to which user, event etc. the data relates etc. End user information needs such metadata, too.	Existence of standard data format and standard API specifications.	Updated	Yes	High	O1, O5	Open	No

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
TMS-5	Secure and robust communication	Public transportation is a critical social infrastructure. Communication among players of the public transportation open data system should be secure and robust.	Existence of cryptographic communication mechanism and fault tolerance mechanism.	Old	Yes	High	O6	Done	Yes
TMS-6	Developers' Site	For the open innovation with public transportation open data, it is necessary to provide an open development environment on the platform called "developers site".	Existence of developers' site of Tokyo Public Transportation Open Data.	Old	Yes	High	O1, O6	Open	No
TMS-7	Data License	Good data license is necessary for the deployment of the data. For the data license, the balance between data holders' merits and users' merits is essentially important.	Existence of open data license.	Old	Yes	High	O1	Open	No
TMS-8	Third party apps	Open data is a method for the collaboration between data holders and application developers.	Number of third party apps or software.	Old	Yes	High	O1, O6	Open.	No
TMS-9	Services for the physically challenged	Contribution to improve quality of life of the physically challenged and the elder in moving in Tokyo is one of the most important objectives.	Number of services for the physically challenged.	Old	Yes	High	O4	Open.	No
TMS-10	User feedback collection	User feedback is necessary for the evaluation of the use case experiment.	Number of user feedbacks.	Old	Yes	High	O6	Open.	No
TMS-11	Visualization of data	The visualization component is the front end for end user that provides them access to interesting data such as the location of vehicles.	Visual data access for users	New	No	High	O3	Done for location on maps.	Yes

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
TMS-12	Data processing	The open data from public transportation operators need to be processed to produce interesting output to the end users (for example, finding the shortest path between location A and B, or the fastest service between two locations, etc.)	Availability of analytical services	New	Yes	medium	O1, O3	Open.	Yes
TMS-13	Control over data collection	This allows users to be empowered to make a decision about what data is collected while the end user is in Tokyo region (such as automatically collected GPS location, , etc.)	UI for controlling use of data	New	Yes	High	O3	Done for certain services	No
TMS-14	Synchronized clocks between IoT devices	Time synchronization is required so that sensor data from different sensors can be correlated by time	Maximum clock drift (usually milliseconds.)	New	No	High	O4	Open.	No
TMS-15	Efficient query of transportation open data	Timely visualization of interesting data requires efficient retrieval of sensor and open data and processed data from sensor streams and data sources.	Query latency in sec.	New	Yes	High	O1	Open.	No
TMS-16	Guarantees for base functionality	When communication between the cloud and the mobile terminal of end users is cut off, cached data can be used to some extent. But the nature of transportation information which requires the always up-to-date status of the transportation service, the loss of communication basically means that the service is degraded very badly.	User value	New	No	Low	O4	Described	No
TMS-17	Data completeness	This actually falls within the responsibility of transportation operators who provide the data to end users.	Percentage of lost packets	New	No	High	O4	Open.	No

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
TMS-18	Data sharing	Data from various data sources need to be shared. End user information needs to be shared with certain limit.	Availability of data sharing mechanisms	New	Yes	High	O1, O3	Open.	Yes
TMS-19	Semantic data integration	The data from various data sources must be integrated and offered in a uniform manner.	Ontologies for data to be shared	New	Yes	High	O5	Due to the partner's request, the open data facility is not deployed in the prototype at this stage.	No
TMS-20	Scale to millions of users	The data sources are expected to be used by a few million users daily to cover the whole Tokyo area private delivery operations.	Number of data points ingested per second, processing latency, storage footprint, and the number of queries	New	Yes	Medium	O6	Open.	No
TMS-21	Communication	If the communication between the cloud servers and the mobile terminal of end users is cut off, then cached information may be usable to some extent. However, due to the nature of transportation data which needs always up-to-date status, such cached data may not be useful all the time.	Number of data queries per seconds, processing latency storage footprint, and required	New	Yes	Medium	O4, O6	Open.	No

5.6.1 Requirements Used for Validation

The following requirements are used for validating the platform. They have been selected for validation as 1) they have been identified as a core requirement that should be provided by the platform, and 2) due to their high priority. Some core requirements could not be validated due to scalability issues and also because those requirements must be satisfied by the transportation operators themselves and not by the CPaaS.io partners.

5.6.1.1 TMS-3 Live Data collection and distribution

This use case deals with real-time operation information of trucks. It is necessary to provide automatic data collection mechanism for these data and provide the data to the users.

5.6.1.2 TMS-5 Secure and robust communication

Communication among stakeholders of the transportation open data system should be secure and robust.

5.6.1.3 TMS-11 Visualization of data

The visualization component is the front end for end user that provides them access to real time photo data and location information on maps. When an end user requests and the partner agrees, it can be also provided to the end user.

5.6.1.4 TMS-12 Data processing

The data from transportation operators need to be processed to produce usable output for optimization. Intelligent processing of photo data to track delivery status is used for validation.

5.6.1.5 TMS-18 Data sharing

Data from various data sources need to be shared. Real time sharing between driver' apps and plan operators are important.

5.6.2 Requirements NOT Used for Validation

The following list of requirements are not used for validation. For each requirement we describe briefly why.

5.6.2.1 TMS-1 Automated data collection mechanism

Automated data collection mechanism. This refers to the data collection done by the transportation operators. This is not done at the initial stage. Evaluation is needed to identify how to facilitate cost-benefit automatic data collection. Although it is very important, this is beyond the control of a CPaaS.io partner in Japan at the initial adoption. Thus, it is excluded from the validation criteria.

5.6.2.2 TMS-2 Multi-lingual data

Multi-lingual data is not needed much in Tokyo Management of Service Vehicle prototype as the partner does not require it at this stage. Therefore, this requirements has been descropped.

5.6.2.3 TMS-4 Data format and API standardization (Metadata + API)

Since smart city transportation is operated by many private companies, data distribution can only be achieved efficiently based on standard data formats and APIs. Sensor data and open data from transportation operators can only be processed if it is known how sensors were deployed, operated, to which user, event etc. the data relates etc. This is very important for data distribution. Due to partner's request, open data format is not deployed and validated in the prototype at this stage.

5.6.2.4 TMS-6 Developers site

For the open innovation with transportation open data, it is necessary to provide an open development environment on the platform called "developers site". It is important, but the partner is not ready for opening the private data, it is excluded from validation at this stage.

5.6.2.5 TMS-7 Open data license

Good data license is necessary for the deployment of the data. For the data license, the balance between data holders' merits and users' merits is important, but the partner is not ready for opening the private data, it is excluded from validation at this stage.

5.6.2.6 TMS-8 Third party apps

Open data is a method for the collaboration between data holders and application developers. So the more number of apps means that the open data infrastructure is more successful. However, the partner is not ready for opening the private data, it is excluded from validation at this stage.

5.6.2.7 TMS-9 Services for the physically challenged

This is important for future cities. However, the provision of the services depend on the data providers, i.e., transportation service operators. Such provision is beyond the control of CPaaS.io partners. Thus it is excluded from the validation criteria.

5.6.2.8 TMS-10 User feedback

At this stage, the transportation operators has just begun to deploy the real time operation assistance. They are not quite ready to listen to the user feedback based on the use of the published open data. So we have excluded this from validation criteria due to their burdens.

5.6.2.9 TMS-13 Control over data collection

This allows users to be empowered to make a decision about what data is collected while the end user is in Tokyo region (such as automatically collected GPS location, the destination of a trip, etc.) This prototype provides an enterprise application and does not support this feature. This requirement is more about the application built on top of CPaaS.io infrastructure and not about the infrastructure itself. Thus, we have excluded this from the validation criteria.

5.6.2.10 TMS-14 Synchronized clocks between IoT devices

Time synchronization is required so that sensor data from different sensors can be correlated by time. However, this prototype does not utilize synchronized operation among operators. These are to be done by public transportation operators, and NOT BY CPaaS.io consortium. Thus, we have excluded this from the validation criteria.

5.6.2.11 TMS-15 Efficient query of transportation open data

Timely visualization of interesting data requires efficient retrieval of sensor and open data and processed data from sensor streams and data sources. Due to the partner's request, query of open data is excluded from this prototype at this stage.

5.6.2.12 TMS-16 Guarantees for base functionality

When communication between the cloud and the mobile terminal of end users is cut off, cached data can be used to some extent. But the nature of transportation information which requires the always up-to-date status of the transportation service, the loss of communication basically means that the service is degraded very badly. Since there is not much we can do in the face of communication cut-off, we descoped this requirement and excluded it to be used for validation.

5.6.2.13 TMS-17 Data completeness

This actually falls within the responsibility of transportation operators who provide the data to end users. Thus, we have excluded this from the validation criteria.

5.6.2.14 TMS-19 Semantic data integration

The data from various data sources must be integrated and offered in a uniform manner. Note that this is primarily the responsibility of the public transportation operators.

5.6.2.15 TMS-20 Scale to millions of users

The data sources are expected to be used by a thousand users daily. So the system as a whole need to support such scalability. The current prototype has estimated to have a capacity of 10-scale expansion. A million-scale users to cover the whole Tokyo delivery is not in scope of this prototype. Due to resource issues, testing at such a scale in the private company context won't be done within CPaaS.io context. Thus, we have excluded this from the validation criteria.

5.6.2.16 TMS-21 Communication

If the communication between the cloud servers and the mobile terminal of end users is cut off, then cached information may be usable to some extent. However, due to the nature of transportation data which needs always up-to-date status, such cached data may not be useful all the time. Since there is not much we can do in the face of communication cut-off, we excluded this from the validation criteria.

6 Waterproof Amsterdam

The rain proof Amsterdam application is a single, but highly relevant application that will use the CPaaS.io platform to improve water management. Through smart, connected rain buffers, water supply peak shaving can be done in order to prevent streets, sidewalks and basements from flooding during heavy rainfall. What makes this application interesting, is the ability to orchestrate the operation of multiple buffers simultaneously across a geographical network. This way, significant impact is expected which will actually address the existing bottlenecks of insufficient water processing capacity.

Update: The use case project has now the approval and the support of Waternet, the local water management company. A start-up under the name De Dakdokters⁶ has joined the project, they are manufacturing green roofs that use the collected rain water to irrigate roof gardens.

6.1 Application description

As research shows, extreme rainfall is becoming fiercer and will occur more frequently. The application is a part of an effort initiated in Amsterdam by Waternet to prevent damage during this kind of weather. By leveraging new smart-roof technologies and rain collectors (rain barrel, underground tanks) along smart city solution, the goal is to create a network of water reservoir that will improve cities' water absorption.

Roughly speaking, a solution has to deal with information from three different areas: the private domain, the public domain and weather forecast. For an optimal contribution of the smart water management system to the existing water management infrastructure, these three domains need to be integrated, featuring an iterative self-learning system which considers the limitations of the existing water management infrastructure. The buffers will be productised for the use in residential and commercial areas. The main premise is that an owner of a smart rain barrel can determine to what extent he / she wants to connect and share on the public network and how permission is given to be overruled by a third party such as Waternet (Amsterdam regional parastatal water management company) in case of predicted extreme rainfall or drought.

To that end, the application will periodically fetch data from a weather platform. It will attempt to use local weather platform with better geographic data and timeline or register to the local weather forecast alerts.

⁶ <http://dakdokters.nl/en/green-roofs/>

In case a potential harmful rain is detected, it will calculate how to setup the water network to avoid damage with the minimum amount of action needed.

6.2 Updated Scope

On the device side we added support for IP devices, i.e., Internet-ready devices. So now a device in charge of collecting rain can either use LoRaWAN or HTTP. The devices will report their status, containing at least the current amount of water held, on a set time and it will be possible to send command messages to the devices to release or retain water.

To match the CPaaS.io objectives, efforts have been directed towards the structure of the application and its security. The structure is now based on micro-services that integrate the different services of CPaaS.io, notably FogFlow and the security layer.

On the user interface side, we decided to change the approach: We won't provide a full device owner dashboard or app, only a simple web app for device owners to visualize their data as a POC. This is motivated by the fact that each manufacturer has a different approach to their user relations and business model. Furthermore, it would be a duplicate of the CPaaS.io dashboard as it would show the same data (all sensor data). The device makers will have access to all their deployed devices that are on the platform with data access defined in the User Agreement they present to their user, so they can integrate the data in their own solution.

We will keep developing a global dashboard for the water management company to aggregate sensor data and display it on a map.

The scope is motivated by the following objectives:

Objective 1: Develop an open social city platform. Smart rooftops are mainly installed on top of buildings that either host offices or apartment. In those cases, occupants might be able to visualize the data reported but more importantly use the platform to build other applications such as an automated garden or redirect the water inside the building.

Objective 2: Deploy the city platform as a service solution. First, data must respect a defined format which avoids different interpretation of data semantics, though it stays free on the choice of the displayed physical unit. Second, a service will be put in place to monitor the device behaviour. It will be able to detect unusual data changes and raise an alert that the device might need intervention.

Objective 3: Empower the citizen and her data. The user is the sole owner of the data and is free to delete any records. Other parties can be given access through a CPaaS.io capability token. User will want to give access to the manufacturer depending on the provided service, and mostly to the city water management company if the user agrees to share the control of its buffer. Being part of the application is a choice that can be changed at any time.

Objective 4: Validate the platform with use cases providing public value. The applications main objective is to reduce structural damages provoked by intense rain. Less public money and workers in the street to repair those damages.

Objective 5: Develop blueprints for adaptations and transfer of solutions to other cities. All the services settings and deployment are defined in a FogFlow service topology. In practice, duplicating the topology and change a few settings will suffice to create a new instance of the solution.

Objective 6: Create impact in cities. The data are formatted and exploitable by any other parties, given they are given access to the data. The smart roof project also has impact in making the city more water efficient. De Dakdokters roofs for example are gardens that will attract new wildlife and help treat pollution.

6.3 Core functionality and component

A first version of the application was realized but it was lacking stability and flexibility. A second version is in the making. It aims for a more scalable architecture using microservices and leverage Fog Flow for deployment. The whole application will be separated in different microservices contained by docker. TTN will host two bridges where the sensors data are published (an HTTP endpoint and an MQTT client connected to TTN), the analytics service, device registry and context are orchestrated by Fog Flow and finally, any authorized dashboard can connect to the Fog Flow context manager. Overall it will improve stability, configuration, scalability, service integration and security.

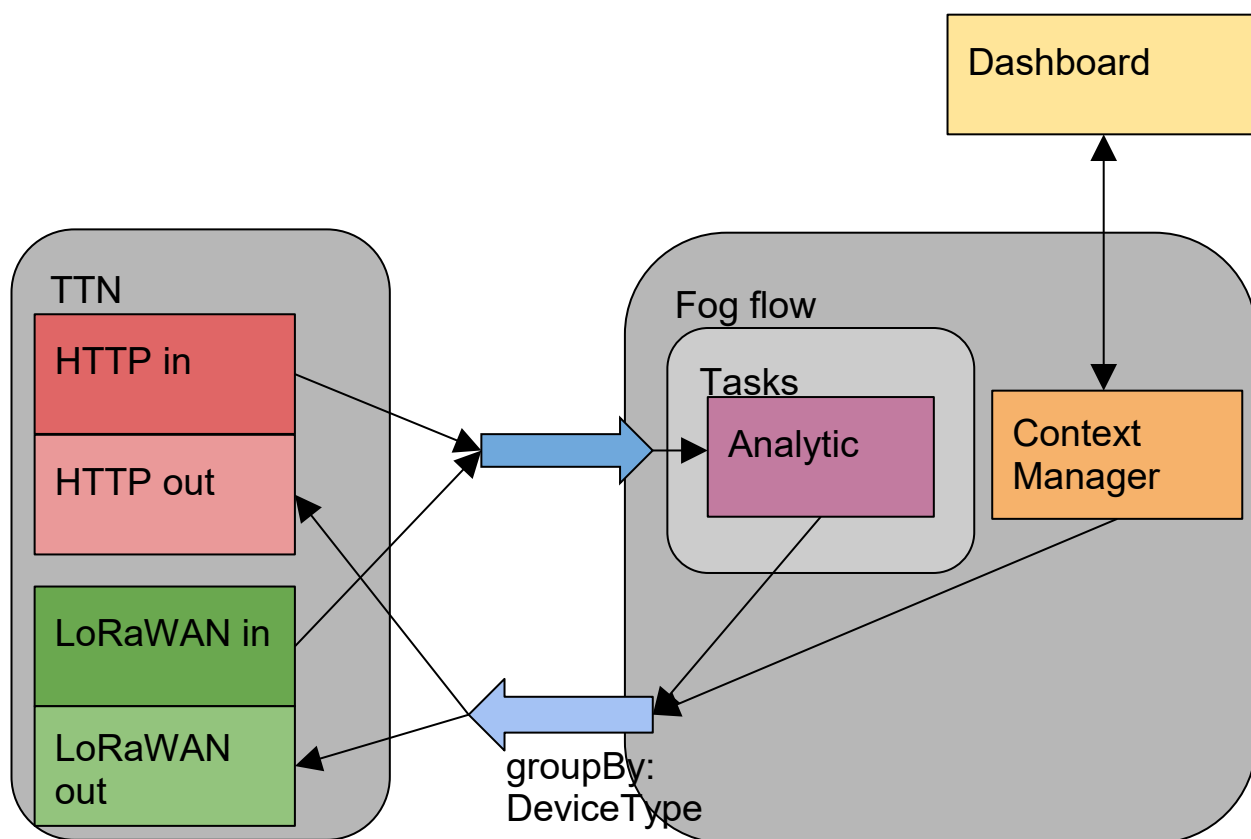


Figure 3 - Waterproof architecture

One of the motivations to use Fog Flow was the ability to create streams of data or messages. We defined two streams, inbound and outbound, all under the same id, but each message has a DeviceType field to differentiate the origin of the data and which bridge to use.

The model is two bound, I/O bounds with two different interfaces:

- Inbound (in):
 - Basic http server where devices can push their sensor data.
 - LoRaWAN, a TTN – Fog Flow integration that converts LoRaWAN uplinks to proper JSON NGSI-10
 - We are considering a websocket bridge to avoid a complex callback setup when using IP capable devices. When installed on a home network, for example, it can be difficult to create such a callback

- Outbound:
 - An HTTP callback is open to listen for messages in the outbound stream having the DeviceType HTTP. The messages are then converted for the device to understand them.
 - Another HTTP callback is open to listen for message in the outbound stream having the DeviceType LoRaWAN. They are then formatted to fit the LoRaWAN requirement.

Other services can subscribe to the generated stream of data, which includes the Waternet analytics service. This service constantly monitors the state of the water buffer in case actions need to be taken. It will also periodically pull a weather report.

6.4 Data set

We use the following types of data, nothing has changed:

1. **Data from the private domain.**
 - a. The type of storage (rooftop, rain barrel, underground storage)
 - b. The per hour uptake and release capacity
 - c. The drainage destination (direct to sewerage, into garden soil
 - i. with varying levels of soil humidity, for instance connected to soil sensors
 - d. Situation of buffers across a city, determining the density of the buffer network
- 7 **Data from the public domain:**
 - a. (Historical) water-related data: known areas of water flooding and other rainfall related issues, ground water levels
 - b. Water infrastructure: sewerage, capacity, bottlenecks, etc.
 - c. Data from weather forecast agencies

Depending on the availability of data to be stored in the CPaaS.io platform, the application may draw from this data to get data on local water infrastructure and weather. All those data are processed by the application (Note: only the water level and debit are processed for the water tank).

Weather data	
Detailed Description	Upcoming weather displaying periods of heavy rain or drought
OGD or private data	OGD
Personal Data	No
Hosting	Platform
Data Provider	KNMI – Dutch weather forecast agency
Format	JSON, XML
Update Frequency	Hourly
Update Size	10kb
Data Source	Sensors
Sensor	Water sensor
Number of Sensors	unknown

Water tank data	
Detailed Description	Filling degree of a water buffer (rooftop, barrel, underground storage); Any other data provided by the device.
OGD or private data	Private
Personal Data	Yes
Hosting	External
Data Provider	TTN / Waternet
Format	JSON
Update Frequency	Hourly or instantly with sudden increasing levels

Update Size	10b
Data Source	Sensors
Sensor	Water sensor
Number of Sensors	1 per buffer

Municipality water infrastructure	
Detailed Description	Geographical data on water infrastructure depicting capacity and ground water levels
OGD or private data	OGD
Personal Data	No
Hosting	Platform
Data Provider	Waternet
Format	XML
Update Frequency	Hourly
Update Size	1kb
Data Source	Sensors, maps
Sensor	Water sensor
Number of Sensors	unknown

7.1 Deployment plan

Over the last year, we made progress on the deployment of smart roofs, thanks to the addition of De Dakdokters. We also defined, prototyped and prove the concept. For the next months we will work toward a first market ready LoRaWAN roof, a connection to the security layer of CPaaS and FogFlow deployment.

Smart roofs have a lot of constrains to be deployed and it is unsure that cities will have the same regulation as Amsterdam, so we are holding off on plans to deploy those roofs in other regions or countries. Nonetheless it is perfectly possible to connect any devices for a proof-of-concept in other regions.

- M01~M12: Requirement analysis and service/system design
 - Align all stakeholders and receive contractual commitment from Waternet for the development of the use case (M04-M05)
 - Investigate areas in the city that dictate the highest urgency or best applicability for a pilot
- M03~M18: System and service development
 - Develop a smart, connected hardware prototype in a lab setting.
 - Integrate prototype on The Things Network backend.
 - Set up a physical LoRa network in the pilot area by placing gateway on rooftop in pilot area
 - First and second release prototype
- M06~M28: Experiment and evaluation
 - Test of first and second release
 - Develop 10 more hardware prototypes to be deployed and tested in real life, outdoor setting.

- Deploy prototypes in existing rain buffer projects in business district in Amsterdam (Zuid area).
- Third release prototype

As of today, Waternet is on board, we have prototyped and demonstrated live both LoRaWAN and HTTP devices. New rooftops have been deployed and more are to come.

7.2 Requirements

Table 3 lists the updated requirements.

Table 3: List of requirements for Waterproof Amsterdam

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
WA-1	Data collection	The application collects and stores data from different sources, including sensors and static cadastral and infrastructural information	Existence of data collection infrastructure	Updated	Yes	High	O1	Done	Yes
WA-2	Data processing	A rain buffer contains a set of standard parameters, such as type, location, capacity. These parameters should be complemented with lifetime usage data, such as times/open close, amount buffered/released, active/inactive, filling level, battery status, etc.	Method for collecting metadata and enriching sensor data	Old	Yes	High	O1	Done	Yes
WA-3	User hierarchies	Different user groups can exert different levels of control over a device, so a hierarchy between groups is established	Definition of user groups, definition of access rights	Updated	Yes	High	O3	Partially done	No
WA-4	Manual control over device	The user or shared user can order the buffer to release or retain water.	UI where users can activate/deactivate the release	Updated	No	Low	O3	Partially done	No
WA-5	Automated control over device	Circumstances need to be defined, based on the various data sources, in which device control for user groups are suspended and delegated to a central user/authority.	Presence of algorithms	Updated	Yes	Medium	O4, O6	Open	Yes
WA-6	Adaptive control levels	The rules and circumstances may automatically trigger control levels as they occur.	Control levels defined	Old	No	Low	O3	Open	No

WA-7	Visualisation of device status	Users can access information via a graphical front-end that displays a record of the rain buffer status, such as filling level.	UI that can be accessed by user.	Updated	No	Low	O3	Partially Done	No
WA-8	Operator control centre	Data can be aggregated per geographical area in order to monitor the status of and trigger actions on a number of devices altogether?	UI for network operator.	Updated	Yes	High	O4, O6O	Open	No
WA-9	Push notifications	Circumstances may trigger communication of predefined messages to users such as push notifications.	Reception of notifications	Old	No	Low	O3	Open	No
WA-10	User feedback collection	Prompts in the application can be used to manually collect user feedback.	Collection of feedback	Old	No	Low	O3	Open	No
WA-11	Data harmonisation	Data collected from the various sources needs to be useable and interpretable by the application	Ontologies for data to be shared	Old	Yes	Medium	O1	Done	Yes
WA-12	Physical network	Data transmission is modulated with LoRa, operated by The Things Network network server. The application needs to be integrated with TTN API	Data received in application server via TTN API	Updated	Yes	High	O1	Done	Yes
WA-13	Multicast-Messaging	Multicast messaging	Bulk downlink messages may be sent to control large quantities of actuators at the same time.	Messages sent from control centre	No	Medium	O1	Partially	No

7.2.1 Requirements Used for Validation

The following requirements are used for validating the platform. They have been selected for validation as 1) they have been identified as a core requirement that should be provided by the platform, and 2) due to their high priority.

7.2.1.1 WA-1

Done. All data are stored and annotated in a broker. This is a basic requirement, data need to be stored and accessible for the operators (user or analytic) to visualize and act based on the data.

7.2.1.2 WA-2

Done. Metadata of the water buffer are required when registering the buffer on the platform. Additional metadata are collected for LoRa devices (radio strength, etc.). Metadata such as the physic unit and the optional metrics are required to provide accurate information and detect defect.

7.2.1.3 WA-5

Open. The service has not been deployed yet. Crossing weather data and the current state of the water tanks is one of the main added value of the application.

7.2.1.4 WA-11

Done. Data are following a define format that enforces the ontology of the application. Some freedom is allowed in the choice of the physical unit. If the message doesn't follow the format it is refused. Attributes names must be harmonised otherwise the analytics cannot recover the data, same as the UI and user.

7.2.1.5 WA-12

Done using a proxy, handling the formatting of the message. Will be replaced by complete NGSI integration being deployed on TTN. Mapping data to the correct format defined by WA-11. Therefore required.

7.2.2 Requirements not Used for Validation

7.2.2.1 WA-3

Partial. Some command can be set as "locking" but we need to see how to integrate this with the capability and security service of CPaaS.io and be synchronised with the device manufacturer. This feature required as this only enforce the decision of the service, so the user cannot remotely control the device via CPaaS. The use case is still valid as this is a simple measure to make the network of water tank more reliable.

7.2.2.1.1 WA-4

Partial. Workflow and API exist but not the UI. The point is to have the analytic pilot the buffer, the user come second in this case.

7.2.2.2 WA-6

Open. Same as WA-5. User can input rules, such as release water when the water level is above a limit. A service will pass over those rules when a new message is received. This is not relevant for validation as it is high level feature to provide comfort to the user, it also most of the time implemented by the manufacturer or resellers in their own application.

7.2.2.3 WA-7

Partial. A first UI was made using MyDevices dashboard. Need to move it to the CPaaS.io dashboard. User will be able to visualize the water level of their buffer, the metadata and the other sensor data sent by the device. UI is secondary for the application, visualization should be made through the CPaaS dashboard.

7.2.2.4 WA-8

Open. Waiting on CPaaS.io dashboard. Same a WA-7 for the validation criteria.

7.2.2.5 WA-9

Open. Low importance and priority. Furthermore, it can only be effective and useful once the WA-4, WA-5- WA-6 are done.

7.2.2.6 WA-10

Open. Now end-user have been involved for now. We rely on feedback made by the manufacturer and Waternet. Feedback form will be integrated in the user UI or via email. User feedback are not necessary for the application to be validated. They define the attraction of the application and leads of improvement.

7.2.2.7 WA-13

Partial. Process and workflow are here but the UI is not present. It is to speed up request and reduce network usage. The others services can simply send command one by one, it then no required.

8 Yokosuka Emergency Medical Care

The *Yokosuka Emergency Medical Care application* is a single, but highly relevant use application that will use the CPaaS.io platform to improve first aid management.

The use case has not changed much from V1 (D2.1) except for a triage system.

8.1 Application Description

See V1 (D2.1) since there have not been many changes. The only notable change is that we no longer actively support a triage system since the medical teams do not use them currently at all.

8.2 Architecture

See V1 (D2.1) since there have not been any relevant changes.

8.3 User Interface

See V1 (D2.1) since there have not been any relevant changes

8.4 Data Sets

The data sets in this use case has been updated since V1 (D2.1) as follows:

Yokosuka Emergency Medical Care Data	
Detailed Description	Emergency Medical Care Sensor data
OGD or private data	Private Data
Personal Data	Some include Personal Data
Hosting	Location Information: Emergency Medical Care Server Camera Image: none (connected directory)
Data Provider	Location Information: Emergency Medical Care Server Camera Image: none (connected directory)
Format	ucR (ucode Relation) format (RDF + ucode in JSON format)
Update Frequency	Real-time Update
Update Size	New real-time data is appended to (old) existing data.

Requirements Specification (V2)

Data Source	Emergency service seam and emergency hospitals of the City of Yokosuka
Sensor	14 ambulances in Yokosuka City and Miura City are equipped with sensors such as cameras (image sensors) and GPSs (location sensors).
Number of Sensors	More than 100 sensors will be used.

8.5 Deployment Plan

We made a significant progress in terms of societal impact. The following is an updated paragraph since V1 (D2.1):

We have collaborated with the Yokosuka City Office and all emergency hospitals in the City of Yokosuka. In April 2017, Fire authorities in Miura city were integrated with one in Yokosuka City, so ambulances in Miura city have been incorporated in our Emergency Medical Care system.

For the next step, we will extend the deployment to the whole Miura Peninsula, a region which includes Yokosuka City, Zushi City, Kamakura City, and Hayama Town. These city and town halls are collaborating, and they often adopt common municipal service systems as a standard. After that, we will try to deploy our system across Japan.

This Yokosuka emergency medical service use case started from the 2nd year of the CPaaS.io project. So, the development and experiment plan are as follows:

- M01~M09: Requirement analysis and service/system design
- M10~M25: System and service development
- M16~M21: Experiment

8.6 Requirements

Table 4 lists the updated requirements. In addition to the original requirements listed in V1 (D2.1), we have looked at the requirements listed for the EU use cases and added appropriate relevant items to the list.

Table 4: Yokosuka Emergency Medical Care Requirements

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
YEM-1	Data collection	This use case deals with real-time operation information of such as patients' conditions, and position of ambulances. It is necessary to provide automatic data collection mechanism for these data.	Existence of automatic collection system	Old	Yes	High	O1	Done	Yes
YEM -2	Consensus for using medical and personal data by city government	The data may have direct or indirect relations with personal information of patients. Before the trial, we need to have formal consensus with city government on the basis of official guidelines for personal data.	Agreement with Yokosuka city government	Old	Yes	High	O1	Done	Yes
YEM -3	Privacy protection of patients	We must provide strong data protection mechanism such as secure communication and secure storage.	Existence of privacy data protection mechanism	Old	Yes	High	O1, O3	Done	Yes
YEM-4	Low-effort for emergency medical team staff members	Emergency medical team staff members are doing very hard work for taking care of patients. It is impossible for them to spend time and effort for the operation of this system. We must realize a good work-flow and user interface for the team staff.	Existence of low-effort mechanism and evaluation score by the team staff members	Old	Yes	High	O1, O3	Done	Yes
YEM-5	User feedback collection	User feedback is necessary for the evaluation of the use case experiment.	Number of user feedbacks	Old	Yes	High	O3	Done	Yes

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
YEM-6	Metadata management	Sensors are used to obtain vital data for patients. Sensor data can only be processed if it is known how sensors were deployed, operated, to which user, event, etc. the data relates, etc.	Method for collecting metadata and enriching sensor data	New	Yes	High	O1	Done	Yes
YEM-7	Visualization of personal data	The visualization component is the front end towards end user (doctors, dispatchers, etc.) that provides them with the up-to-date health related vital data of the patient.	Visual data access for end user	New	No	Medium	O1, O3	Done	No
YEM-8	Data processing	Emergency medical system does not need this much as of now in the current scenario/form.	Availability of analytical services	New	Yes	low	O1, O3	Done	No
YEM-9	Control over data collection	This allows patients to be empowered to make a decision between what data is collected while the patient is in the ambulance.	UI for controlling use of data PATIENTS CAN REFUSE to be monitored using the remote diagnosis, etc. BUT BASICALLY, THIS MEANS THE PATIENT DOES NOT USE EMERGENCY MEDICAL SERVICE EFFECTIVELY.	New	Yes	Medium	O3	Done	No
YEM-10	Synchronized clocks between IoT devices	Time synchronization is required so that sensor data from different sensors can be correlated by time	Maximum clock drift (usually minute order suffices for bed assignment purposes, but milliseconds resolution is necessary for medical devices to have correlation of vital data such as heart beat.)	New	No	High	O4	Done	No

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
YEM-11	Efficient query of patient, hospital bed and doctor data	Personalized visualizations require efficient retrieval of media data and processed data from sensor streams. Finding unused bed at hospitals where doctors with appropriate specialization are available require quick search of such data.	Query latency in sec.	New	Yes	High	O1, O3	Done	No
YEM-12	Guarantees for base functionality	When communication between the ambulance and the hospitals is lost, the system downgrades to the traditional phone-based services.	User value	New	No	Low	O6	Done	No
YEM-13	Data completeness	YEM does not need the fail over *IF* the communication between the sensor devices and collection node is lost. Basically, the system downgrades to the traditional service WITHOUT ACCESS to vital data for the remote diagnosis.	Percentage of lost packets	New	No	High	O6	Done	No
YEM-14	Data sharing	Patients want their medical data shared between doctors at various hospitals. Also, the status of beds and doctors need to be shared among the hospitals and dispatcher's office.	Availability of data sharing mechanisms	New	Yes	Low	O1, O3	Done partially	Yes

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for Validation
YEM-15	Semantic data integration	The collected data inside ambulance is currently fixed data. Bed and doctor availability is also fixed information. But, in general, data sharing between diverse systems can only efficiently be achieved if the semantics of the data are machine readable	Ontologies for data to be shared	New	Yes	Low	O4	Done	No
YEM-16	Scale to hundreds of thousands of people	The service is offered right now per a district that services hundreds of thousands of people. However, do note that the number of emergency patients at a time is NOT that large. The numbers of beds are in the order of tens of thousands and the doctors are in the range of thousands at most for now. The platform needs to be able to efficiently ingest, store, process user data and hospital bed and doctor status.	Number of data points ingested per second, processing latency, storage footprint	New	Yes	Medium	O6	Done	No
YEM-17	Communication	The application collects medical data of the patient inside the ambulance and share this via communication to doctors at various hospitals. The platform needs to be able to efficiently ingest, store, process patient data.	Number of data points ingested per second, processing latency, storage footprint	Old	Yes	Medium	O6	Done	Yes

8.6.1 Requirements Used for Validation

The following requirements are used for validating the platform. They have been selected for validation as 1) they have been identified as a core requirement that should be provided by the platform, and 2) due to their high priority. Some items are not used for validation due to reasons cited below.

8.6.1.1 YEM-1 Data collection.

This use case deals with real-time operation information of such as patients' conditions, and position of ambulances. It is necessary to provide automatic data collection mechanism for these data.

8.6.1.2 YEM-2 Consensus for using medical and personal data by city government

The data may have direct or indirect relations with personal information of patients. Before the trial, we need to have formal consensus with city government on the basis of official guidelines for personal data.

8.6.1.3 YEM-3 Privacy protection of patient

We must provide strong data protection mechanism such as secure communication and secure storage.

8.6.1.4 YEM-4 Low-effort for emergency medical team staff members

Emergency medical team staff members are doing very hard work for taking care of patients. It is impossible for them to spend time and effort for the operation of this system. We must realize a good workflow and user interface for the team staff. This is important for the system to be accepted by the working staff members.

8.6.1.5 YEM-5 User feedback collection

User feedback is necessary for the evaluation of the use case.

8.6.1.6 YEM-6 Metadata management

Sensors are used to obtain vital data for patients. Sensor data can only be processed if it is known how sensors were deployed, operated, to which user, event etc. the data relates, etc.

8.6.1.7 YEM-14 Data sharing

Patients want their medical data shared between doctors at various hospitals. Also, the status of beds and doctors need to be shared among the hospitals and dispatcher's office for emergency medical team staff members to select an acceptable hospital. However, in Yokosuka City, hospitals cooperating in this use case almost accept patients. Thus, in this use case, only their medical data was used for validation.

8.6.1.8 YEM-17 Communication

The application collects medical data of the patient inside the ambulance and share this via communication to doctors at various hospitals. The platform needs to be able to efficiently ingest and process patient data.

8.6.2 Requirements NOT Used for Validation

8.6.2.1 YEM-7 Visualization of personal data

Visualization component is the front end towards end user (doctors, dispatchers, etc.) that provides them with the up-to-date health related vital data of the patient. Except for the visual capture of the patient using a camera, the visualization of data is not high priority.

8.6.2.2 YEM-8 Data processing

Emergency medical system does not need this much as of now in the current scenario/form.

8.6.2.3 YEM-9 Control over data collection

This is not used for evaluation due to the following reason: patients can make a decision between what data is collected while the patient is in the ambulance.

Requirements Specification (V2)

PATIENTS CAN REFUSE to be monitored using the remote diagnosis, etc. BUT BASICALLY, THIS MEANS THE PATIENT DOES NOT USE EMERGENCY MEDICAL SERVICE EFFECTIVELY. The patient, if conscious, can refuse the data collection. But so far, none has done so.

8.6.2.4 YEM-10 Synchronized clocks between IoT devices

Time synchronization is required so that sensor data from different sensors can be correlated by time. Usually minute order accuracy suffices for bed assignment purposes, but milliseconds resolution is necessary for medical devices to have correlation of vital data such as heart beat. However, as of now, no accurate measurement regarding to timestamp is necessary for the type of vital data we collect.

8.6.2.5 YEM-11 Efficient query of patient, hospital bed and doctor data

Personalized visualizations require efficient retrieval of media data and processed data from sensor streams. Finding unused bed at hospitals where doctors with appropriate specialization are available require quick search of such data. Infrastructure was put in place, but the data from the hospitals and doctors were not made available so far. This is beyond the control of CPaaS.io partners. Thus, this was not used for validation.

8.6.2.6 YEM-12 Guarantees for base functionality

When communication between the ambulance and the hospitals is lost, the system downgrades to the traditional phone-based service. This is the default operation mode.

8.6.2.7 YEM-13 Data completeness

YEM does not need the fail over *IF* the communication between the sensor devices and collection node is lost. Basically, the system downgrades to the traditional service WITHOUT ACCESS to vital data for the remote diagnosis.

8.6.2.8 YEM-15 Semantic data integration

The collected data inside ambulance is currently fixed data. Bed and doctor availability are also fixed information. But, in general, data sharing between diverse systems can only efficiently be achieved if the semantics of the data are machine readable. But due to the relatively small fixed data set, we do not use this as validation factor.

8.6.2.9 YEM-16 Scale to hundreds of thousands of people

The service is offered right now per a district that services hundreds of thousands of people. However, do note that the number of emergency patients at a time is NOT that large. The numbers of beds are in the order of tens of thousands and the doctors are in the range of thousands at most for now. The platform needs to be able to efficiently ingest, store, process user data and hospital bed and doctor status.

9 Smart Parking

In this deliverable we are introducing a new use case related with one of the most common issues that big cities cope with daily, traffic congestion. Helping citizens in the task of finding a parking spot is therefore an important contribution to reduce the amount of time that vehicles move along the city.

Murcia is a medium-size city in the south east of Spain with a population of 450.000 residents. This number increases by about 5.000 residents per year, with a noteworthy distribution of people among the city districts. Because of this and the fact that Murcia is the capital of the Murcia Region, it has seen a dramatic rise of accesses to the city centre in the last years. Day by day, commuters, tourists and families traveling by car collapse the core of Murcia, wanting to park at commerce, financial and historical areas.

Requirements Specification (V2)

To improve the situation, the aim of this use case is that of taking advantage of the CPaaS.io platform to provide a service that tracks the state of the parking spots to provide the drivers with this information beforehand, so they can better plan where they will park. This will also lead to a more fluid traffic in the centre of the city, because there will be less drivers wandering, looking for a parking spot in areas that do not have any available.

9.1 Application Description

Our objective is to provide an easy-to-use app which will allow the user to obtain an indication of the most probable area to look for a free parking spot closer to her desired destination. To do so, this app makes use of the framework called FogFlow which is detailed in D4.1 and D4.2.

9.2 Core Functionalities and Components

According to the requirements specified in D2.1 we provide the following architecture. For this use case the platform provides the data and their analytics capabilities in such a way that The Arrival Time Estimation module will be able to extract the information and provide the user with the best area to find a free parking spot.

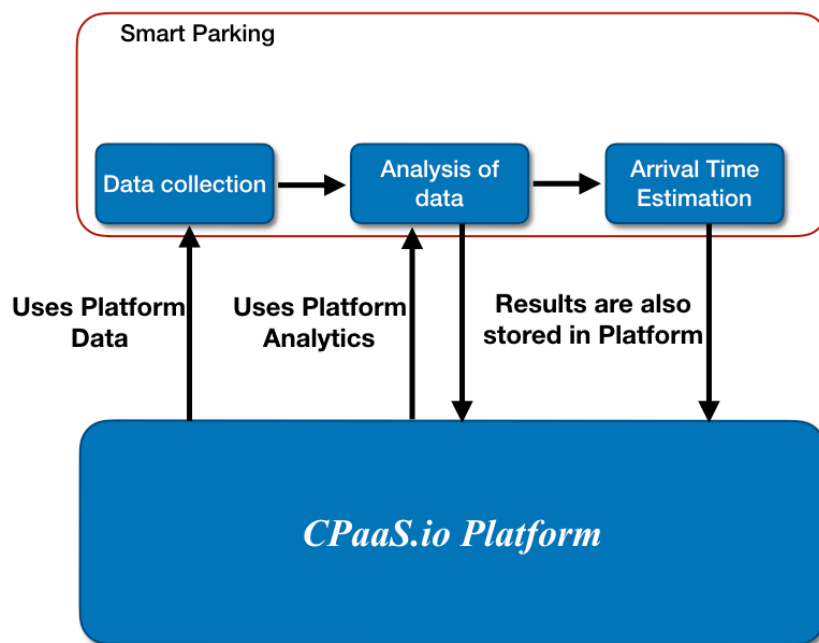


Figure 4. Diagram of the architecture for Smart Parking

The following diagram, Figure 5, presents the different tasks comprising this use case, as well as the flow of information among them.

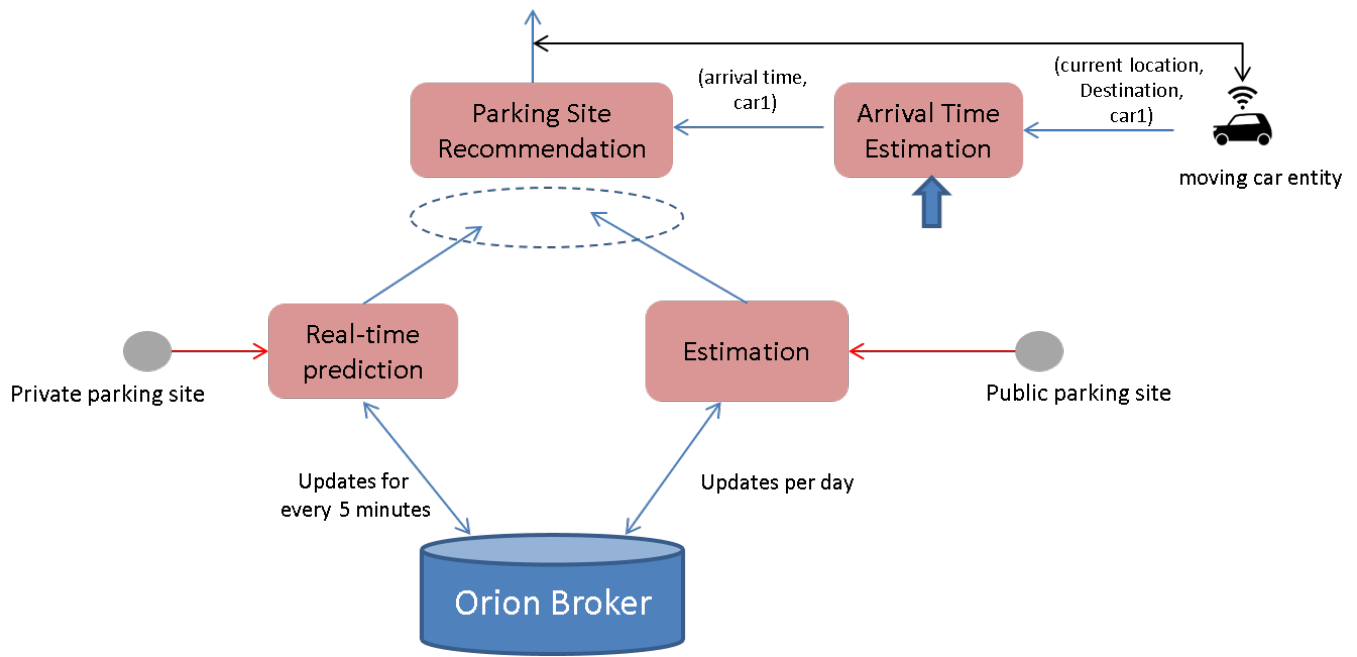


Figure 5: Diagram of the tasks for the use case

Following a bottom-up approach, we can see the following tasks:

Real-time prediction task: This task corresponds to the private parking sites; whose data is available in real time. The town will be divided in five zones, each having multiple private parking sites. A task instance will be launched for each zone, listening to occupation changes from its parking sites, but will only publish significant changes, i.e., only when the difference in available spots is above a certain threshold, because a change in a single parking spot will not change the behaviour of the system.

Estimation task: This task corresponds to the regulated parking zones of Murcia city. Their data is received by the system once a day, containing all the tickets issued that day, and therefore the data is not available in real time. Because of that, an intelligent application will predict the occupation based on the previous data. However, the prediction should be updated daily to include the new data. This task will read the new data, update the prediction application and then publish it so the other components can use it.

Parking Site Recommendation task: The results from the two previous tasks will feed this task, which will be used to generate the response the user will receive.

Arrival Time Estimation task: This task is the endpoint receiving the requests by the users which are parametrized by specifying the location of the vehicle, as well as the desired destination.

This information is extended in D4.1.

9.3 Data Sets

Since this use case is new, in this section we detail the different involved datasets. The application needs the information provided by the concessionary companies responsible for the management of private parking sites and the regulated parking zones (RPZs). This information is therefore divided in two groups.

On the one hand, the information regarding private parking sites:

- Id: An identifier of the parking site

Requirements Specification (V2)

- Type: The current value for this entity is *Sensor*
- Attributes:
 - o name: The name of the parking site
 - o geolocation: Geographical position of the parking site
 - o forced: This attribute can be set to establish a number of free parking spots publicly available.
 - o free parking spots: The number of free parking spots.
 - o total parking spots: The number of total parking spots.

Such information will be transmitted using NGSI the following way:

```
{
  "contextResponses": [
    {
      "contextElement": {
        "attributes": [
          {
            "name": "forzado",
            "type": "number",
            "value": "0"
          },
          {
            "metadata": [
              {
                "name": "location",
                "type": "string",
                "value": "WGS84"
              }
            ],
            "name": "geoposicion",
            "type": "coords",
            "value": "37.9942698,-1.1294058"
          },
          {
            "name": "libres",
            "type": "number",
            "value": "113"
          },
          {
            "name": "nombre",
            "type": "number",
            "value": "Morales Meseguer"
          },
          {
            "name": "plazas%20libres",
            "type": "number",
            "value": "118"
          },
          {
            "name": "plazas%20totales",
            "type": "number",
            "value": "220"
          },
          {
            "name": "totales",
            "type": "number",
            "value": "220"
          }
        ],
        "id": "Aparcamiento:105",
        "isPattern": "false",
        "type": "Sensor"
      },
      "statusCode": {
        "code": "200",
        "reasonPhrase": "OK"
      }
    }
  ]
}
```

Requirements Specification (V2)

Detailed Description	Availability of 5 private parking sites. The information contains both total amount of parking spots and the free ones
OGD or private data	Private
Personal Data	No
Hosting	Platform
Data Provider	Murtrafic – Dutch weather forecast agency
Format	JSON, XML
Update Frequency	Every 5 min
Update Size	--
Data Source	Concesionary Company platform
Sensor	--
Number of Sensors	unknown

On the other hand, regarding the RPZs information we have received in our platform the information about the tickets issued the day before. Such information comprises:

- id: identifier of the ticket, e.g. Ticket:XXXXXXX.
- type: for these entities the value is set to Ticket.
- Attributes:
 - o duration: duration in seconds.
 - o expender: expender
 - o date_of_issue: issued date.
 - o hour_of_issue: issued time.
 - o price: price
 - o payment_method: Payment option (e.g. Efectivo = Cash)
 - o rate: rate
 - o validity: booked period of time e.g.12:09 – 12:54
 - o zone: Sector or area.

Below, you can find an example of the NGSI information per ticket.

```

"contextResponses": [
  {
    "contextElement": {
      "attributes": [
        {"name": "duration", "type": "Integer", "value": 2700},
        {"name": "expender", "type": "Integer", "value": 78},
        {"name": "date_of_issue", "type": "Date", "value": "01/03/2018"},
        {"name": "hour_of_issue", "type": "Time", "value": "12:09"},
        {"name": "price", "type": "String", "value": "0,3 EUR"},
        {"name": "payment_method", "type": "String", "value": "Cash"},
        {"name": "rate", "type": "String", "value": "Azul"},
        {"name": "validity", "type": "String", "value": "12:09 - 12:54"},
        {"name": "zone", "type": "String", "value": "Sector 4"}
      ],
      "id": "Ticket:443994176",
      "isPattern": "false",
      "type": "Ticket"
    }
  }
]

```

RPZ tickets	
Detailed Description	Detailed information about the tickets issued by the expenders. This information comprises, duration, expender, date of issue, hour of issue, value, payment method, rate, validity, zone

Requirements Specification (V2)

OGD or private data	Private
Personal Data	No
Hosting	Platform
Data Provider	EYSA – Spanish concessionary company
Format	JSON, XML
Update Frequency	Daily
Update Size	Dynamic
Data Source	Expenders
Sensor	--
Number of Sensors	unknown

For this scenario is also fundamental the destination where the user intends to park his car.

9.4 Deployment Plan

For this use case, the deployment plan involves the following phases:

1. M18~M20: Deployment of the FogFlow framework
2. M20~M22: Integrate the instantiated components with real information about Murcia city regarding private parking sites and RPZs.
3. M21~M24: Definition of the different tasks and workers responsible for the processing activities achieved in the edge and cloud layer.
4. M23~M26: Development of the application
5. M24~M26: Application Validation

9.5 Requirements

Table 4 lists the updated requirements.

Table 5: Requirements for Smart Parking

ID	Description	Rationale	Fit Criterion	Old/New/Updated	Core req ?	Priority	Rel. Project Objectives	Status	Used for validation
SP-1	Data collection infrastructure	Private parking site info must be provided in real time. RPZ info is provided at the end of the day	Existence of a data collection infrastructure	New	Yes	High	O1, O2, O3, O4, O6	Open	Yes
SP-2	Edge processing	In the edge, processing capability is needed to carry out specific tasks.	Existence of devices able to handle processing tasks in the edge.	New	Yes	High	O1, O2, O3, O4, O6	Open	Yes
SP-3	Cloud processing	In the same sense, complex tasks should be achieved by virtualized services in the cloud	Cloud services able to process complex tasks	New	Yes	High	O1, O2, O3, O4, O6	Open	Yes
SP-4	Communication	Connectivity between different players of the FogFlow framework must be guaranteed	Online availability of services and devices	New	Yes	High	O1, O2, O4, O6	Done	Yes
SP-5	Virtualized services	Fogflow requires the deployment of virtualized services in a specific server	Support for virtualized services	New	Yes	High	O1, O2, O4, O6	Done	Yes

9.5.1 Requirements Used for Validation

The following requirements are used for validating the platform. They have been selected for validation as 1) they have been identified as a core requirement that should be provided by the platform, and 2) due to their high priority.

9.5.1.1 SP-1

Done. All data are stored and annotated in a broker. This is a basic requirement, data need to be stored and accessible for the operators (user or analytic) to visualize and act based on the data.

9.5.1.2 SP-2

Done. Thanks to the FogFlow framework, edge nodes have been deployed according to private parking availability and Regulated Parking Zone daily logs.

9.5.1.3 SP-3

Done. The Recommendation module is stored in the cloud to provide a recommendation to an incoming parking area request for a connected car.

9.5.1.4 SP-4

Done. The components involved in the Smart Parking use case have been deployed using the FogFlow framework. It also guaranteed the communication between them so as to allow a connected car to receive the appropriate recommendation decision.

9.5.1.5 SP-5

Done. Again, thanks to the FogFlow framework, edge and cloud tasks have been deployed using virtualized services.

10 Summary

In this deliverable we have provided an update of the requirements of the applications based on which we develop the CPaaS.io platform. As the focus in the last phase of the project, is on transferability and federation, we have revisited the requirement in particular with respect to covering cross-border aspects that drive the federation aspects of the European and Japanese platform instances. This has for instance be reflected by the requirement EUE-11 (Semantic data integration) and newly added requirement EUE-13 (Access data from different platform instances). We also described the requirements of Smart Parking as an additional CPaaS.io application. In section 2 we have described the new focus of the Enhanced User Experience (MyEvents) Application that should be able to use the data collected at the Sapporo Snow Festival 2018 stored in the Japanese CPaaS.io instance.

While the core of the requirements has been unchanged, some requirements have been updated and added in order to reflect the experiences gained while developing the applications, receiving feedback from stakeholders and discussing with representatives of the technical work packages within the project. We have taken particular care to reflect scalability and privacy requirements into account and link the requirements to the project objectives.

The respective application requirements have been carefully reviewed to identify core requirements, i.e. requirements that developers or other application stakeholders consider to be fulfilled by the platform and possibly be reused by other applications. Furthermore, the requirements have been linked to the project objectives and requirements used for validating the platform have been identified.

Appendix A – Description of the fields used in the data set description

Table 6: Description of the fields used in the data set descriptions

Data Set Name	A name that identifies the data set
Detailed Description	A verbal description of the data set.
OGD or private data	Classifies the data whether it is open governmental data (OGD) or private data, i.e. data that is not provided by governmental institutions. Possible values are <i>OGD</i> or <i>Private data</i> .
Personal Data	Indicates whether information contained in the data set identifies or could be used to identify any individual person. Possible values are <i>yes</i> and <i>no</i> .
Hosting	Indicates whether the data set is planned to be hosted by the CPaaS.io platform or not. Possible values are <i>platform</i> and <i>external</i> .
Data Provider	Describes the entity that is offering the data set. This can be an open data portal, a partner of the consortium or any other private and public entity that provides the data. Possible values include a string that identifies the data provider such as partner name, city name/department and may include a URL.
Format	Describes the technical format in which the data is provided. Possible values include a string referencing a technical format. Examples include CSV for comma separate value or RDF for a serialization format for data represented in the Resource Description Format.
Update Frequency	Describes the rate in which updates of the data sets are available. This may be specified in Hz or by specifying a time interval between an average update, e.g. 1 update per hour, minute or year etc.
Update Size	Describe how big a single update. Possible values include bytes, Kilobytes, etc. or a number of atomic data units such as a record, document, or triple.
Data Source	Technically characterizes the source. Possible values include but are not limited to <i>SQL database</i> , <i>sensor</i> , <i>file</i> .
Sensor	Describes the sensor that provides the data if applicable. Possible values are <i>N/A</i> if the data source is not a sensor or a string describing the sensor including but not limited to <i>wristband</i> , <i>mobile phone</i> ,
Number of Sensors	The amount of sensors used in a deployment if applicable. The field may be extended to specify a finer deployment granularity, e.g. <i>number of sensors per person</i> . Possible values are integer values.

Appendix B –Description of the fields in the requirements tables

Table 2 describes the fields used for the requirements. The ID, Description, Rationale and Fit Criterion are the same as in D2.1.

Table 7: Field description of backlog items

FIELD	DESCRIPTION
ID	Each requirement is uniquely identified of the form <application_id>-<number>, e.g. for requirement 1 of the enhanced user experience application: EUE-1
Description	The description is the intent of the requirement. It is a statement about what the system has to fulfil according to the rationale.
Rationale	The rationale is the reason behind the requirement's existence. It explains why the requirement is important and how it contributes to the system's purpose.
Fit Criterion	A quantification or measurement to assess to which extent the original requirement has been eventually implemented/taken into account within the system. In some cases only the criteria used for quantification may be provided.
Old/New/Updated	Is this an old requirement that has already been described in D2.1 in the same way, or is it an updated requirement that already existed in D2.1 and one of the fields above have been updated, or is it a new requirement that did not exist in D2.1
Core requirement ?	Could this application requirement be a common, application independent requirement for the platform
Priority	Is it a high, medium or low priority requirement
Related Project Objectives	Links to the project objectives Objective 1 (O1): Develop an Open Social City Platform Objective 2 (O2): Deploy the City Platform as a Service Solution Objective 3 (O3): Empower the citizen to her data Objective 4 (O4): Validate the platform with use cases providing public value Objective 5 (O5): Develop blue prints for the adaptation and transfer of solutions to other cities Objective 6 (O6): Create impact in cities
Status	Is this requirement already addressed, is it still open or has it been descoped because it is not a requirement any more.
Used for validation	Is this requirement used for validating the CPaaS.io platform ?